

Longitudinal transcriptional changes reveal genes from the natural killer cell-mediated cytotoxicity pathway as critical players underlying COVID-19 progression

Reviewed Preprint

v2 • July 22, 2024

Revised by authors

Reviewed Preprint

v1 • February 2, 2024

Matias A Medina, Francisco Fuentes-Villalobos, Claudio Quevedo, Felipe Aguilera, Raul Riquelme, Maria Luisa Rioseco, Sebastian Barria, Yazmin Pinos, Mario Calvo, Ian Burbulis, Camila Kossack, Raymond A Alvarez, COVID 19 South Chile Group, Jose Luis Garrido , Maria Ines Barria 

Facultad de Medicina y Ciencia, Universidad San Sebastián, Puerto Montt, Chile • Departamento de Microbiología, Facultad de Ciencias Biológicas, Universidad de Concepción, Concepción, Chile • Departamento de Bioquímica y Biología Molecular, Facultad de Ciencias Biológicas, Universidad de Concepción, Concepción, Chile • Hospital Dr. Eduardo Schütz Schroeder, Puerto Montt, Chile • Hospital Base San José, Osorno, Chile • Instituto de Medicina, Facultad de Medicina, Universidad Austral, Valdivia, Chile • Division of Infectious Diseases, Department of Medicine, Immunology Institute, Icahn School of Medicine at Mount Sinai, New York, NY 10029, USA

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Abstract

Patients present a wide range of clinical severities in response SARS-CoV-2 infection, but the underlying molecular and cellular reasons why clinical outcomes vary so greatly within the population remains unknown. Here, we report that negative clinical outcomes in severely ill patients were associated with divergent RNA transcriptome profiles in peripheral immune cells compared with mild cases during the first weeks after disease onset. Protein-protein interaction analysis indicated that early-responding cytotoxic NK cells were associated with an effective clearance of the virus and a less severe outcome. This innate immune response was associated with the activation of select cytokine-cytokine receptor pathways and robust Th1/Th2 cell differentiation profiles. In contrast, severely ill patients exhibited a dysregulation between innate and adaptive responses affiliated with divergent Th1/Th2 profiles and negative outcomes. This knowledge forms the basis of clinical triage that may be used to preemptively detect high-risk patients before life-threatening outcomes ensue.

Highlights

- Mild COVID-19 patients displayed an early transcriptional commitment with NK cell function, whereas severe patients do so with neutrophil function.
- The identified co-expressed genes give insights into a coordinated transcriptional program of NK cell cytotoxic activity being associated with mild patients.
- Key checkpoints of NK cell cytotoxicity that were enriched in mild patients include: *KLRD1*, *CD247*, and *IFNG*.

- The early innate immune response related to NK cells connects with the Th1/Th2 adaptive immune responses, supporting their relevance in COVID-19 progression.

eLife assessment

This **valuable** paper compares blood gene signature responses between small cohorts of individuals with mild and severe COVID-19. The authors provide **solid** evidence for distinct transcriptional profiles during early COVID-19 infections that may be predictive of severity, within the limitations of studying human patients displaying heterogeneity in infection timelines and limited cohort size.

<https://doi.org/10.7554/eLife.94242.2.sa3>

Introduction

The severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) promotes several dysfunctions in human immune responses while triggering a broad spectrum of clinical presentations that range from asymptomatic infection to a mild, moderate, or sometimes lethal severe symptomatology (Ge et al., 2020 [↗](#); The, 2020 [↗](#); Wu & McGoogan, 2020 [↗](#)). Convalescent patients report prolonged COVID-19 symptoms beyond the time course for typical cold and flu events, which highlights the possibility of long-term tissue damage generated by this virus (Ladds et al., 2020 [↗](#); Nalbandian et al., 2021 [↗](#); Ryan et al., 2022 [↗](#); Subramanian et al., 2022 [↗](#)). Similarly as other respiratory viruses, it is known that SARS-CoV-2 triggers an immune response involving the recruitment, activation, and differentiation of innate and adaptive immune cells (Newton, Cardani, & Braciale, 2016 [↗](#); Shen et al., 2023 [↗](#); Wauters et al., 2021 [↗](#)). For mildly-ill patients, these coordinated immunological efforts resolve infection but for unknown reasons the virus evades these responses in severely-ill patients to produce life-threatening COVID-19 (Park, 2020 [↗](#); Rashid et al., 2022 [↗](#); Sun, Xie, Bu, Zhong, & Zeng, 2022 [↗](#); Thorne et al., 2022 [↗](#)). The genetic background and physiological health of individual patients certainly plays a major role in the clinical presentation of COVID-19 but the exact mechanisms of how the virus evades innate and adaptive responses is not known, or why there is such great variability in severity of clinical presentation among patients. This information is critical for developing new diagnostics that detect patients who will eventually progress to severe COVID-19 before respiratory failure ensues, and furthermore, provide host and virus targets to engineer effective treatments (X. Li et al., 2021 [↗](#); Samadzadeh et al., 2021 [↗](#)).

Biomarkers linked to COVID-19 severity hold promise for detecting patients that will eventually develop severe COVID-19 (Janssen et al., 2021 [↗](#); The, 2020 [↗](#)). In this context, blood-derived cues were associated with severe COVID-19, including an imbalance in immune cell populations that included neutrophil abundance, lymphopenia, myeloid dysfunction, and T cell activation/exhaustion (Ahern et al., 2022 [↗](#); Chen & John Wherry, 2020 [↗](#); Mann et al., 2020 [↗](#); Wauters et al., 2021 [↗](#)). The differential expression of select chemokines and their receptors (Khalil, Elemam, & Maghazachi, 2021 [↗](#)) with associated cytokine storm drove monocyte and megakaryocyte dysfunction in severely ill patients (Ren et al., 2021 [↗](#)). Comprehensive knowledge of host immune responses against SARS-CoV-2 is still limited but these divergent cell profiles implicated cell-to-cell signaling events occurring between the innate and adaptive cell compartments as critical for the progression of severe COVID-19 (Daamen et al., 2021 [↗](#); Rabaan et al., 2022 [↗](#); C. Wang et al., 2022 [↗](#)).

One approach to identifying changes in immune-responses affiliated with severe COVID-19 is to monitor autocrine, paracrine and endocrine signaling in individual patients over time. Temporal events associated with each type of signaling is obviously difficult to disentangle from measuring the activities of circulating peripheral cells alone because there are distinct events happening in localized microenvironments, e.g., the spleen and lymph nodes. A complementary tactic to access information about these events is to monitor gene expression for the synthesis of chemokines and cell-associated receptors as a proxy of biochemical events happening in distinct immune effector cells. Based on current knowledge, we hypothesized that critical events occurring at the earliest stages of infection necessary for effective viral clearance are either perturbed or disrupted so as to promote cytokine storm and other pathologies associated with severe outcomes. We predicted these pathological immune events may be observable by measuring changes gene expression reflecting activities in distinct effector cells during the first weeks of infection (Ahern et al., 2022 [DOI](#); Bernardes et al., 2020 [DOI](#); Notarbartolo et al., 2021 [DOI](#); Xiong et al., 2020 [DOI](#); Zheng et al., 2020 [DOI](#)). However, these types of experiments require careful design because the type and quantity of all immune responses are dynamic during infections and comparing poorly-matched PBMCs may confound identification of bonafide immune dysregulation evident between patients (Bernardes et al., 2020 [DOI](#); Notarbartolo et al., 2021 [DOI](#); Zheng et al., 2020 [DOI](#)).

Here, we designed a longitudinal comparison between mild and severe patients, choosing the appropriate samples according to the clinical progression and the unbiased gene expression profile to study how changes in gene expression in distinct immune effector cells changed during the earliest time points since peak of symptoms and during progression of clinical disease. We repeatedly measured whole-transcriptome profiles of peripheral blood mononuclear cells (PBMCs) from the same cohort of mildly- and severely-ill patients to identify molecular pathways that were enriched during the clinical trajectory of COVID-19 over time. Briefly, to gain more insights into our findings and get more comprehensive knowledge of the phenomenon, we used a pairwise comparison of gene expression, gene set enrichment, and weight-correlated gene network analyses. By doing so, we identified pathways of genes involved with the NK cell cytotoxicity enriched in mild patients when compared to severe. Besides focusing on a particular molecular pathway, we investigated the interactions to better comprehend the underlying phenomena of a successful immune response, contributing to an integrated point of view throughout the transcriptomic analyses of functional pathways to mitigate potential biases attributed to focusing the study on a single pathway. In this regard, we revealed that the NK signaling pathway was intricately related to other transcriptional circuits, such as those governing Th1/Th2 cell differentiation and cytokine-cytokine receptor signaling pathways. These interactions highlight the importance of these pathways as bridges between the innate and adaptive immune responses throughout the disease, implying that the innate NK signaling pathway (cell cytotoxic activity) is beneficial, and possibly a critical activity required to effectively eradicate coronavirus. We also observed that an adaptive immune response including early cell-mediated immunity was significant in lowering disease severity. The link between the primary innate NK cell activity and the transcriptional priming of adaptive Th1 and Th2 cell responses appears to be more robust in mild patients than in severe. This work provides clear guidance to develop better medical practices and prevention tactics against SARS-CoV-2 and other related infectious respiratory virus (Haitao et al., 2020 [DOI](#); Ponti, Maccaferri, Ruini, Tomasi, & Ozben, 2020 [DOI](#); L. Zhang & Guo, 2020 [DOI](#)).

Results

Clinical features and temporal gene expression patterns in SARS-CoV-2 infected patients

A total of 22 peripheral blood samples were obtained from eight COVID-19 patients. These samplings following a longitudinal schedule complemented with 2 samples from healthy donors. All patients were recruited after an average period of 5 days after symptoms onset (**Figure 1A** [DOI](#)).

Some samples were taken from patients at the Hospital of Osorno and the Hospital of Puerto Montt, which are cities located in the region Los Lagos. The remaining samples were collected from patients at the Hospital Base and Clínica Alemana in Valdivia, a city located in the Region Los Ríos. All infections occurred between November 2020 and May 2021 (**Table 1** [↗](#)).

A crucial issue with longitudinal studies is defining an appropriate sampling schedule that provides a reasonable comparison between patients during the time course of naturally occurred infections. To align the comparability and consistency of data measured between patients, we designed a protocol consisting of three donations per patient to monitor events occurring during both acute infection and the recovery phase. We collected peripheral blood samples on days 0, 7, and 28 (D0, D7, and D28) during the peaks of symptoms (**Figure 1B** [↗](#)).

Clinical features of COVID-19 patients with mild and severe symptoms were determined by medical personnel at the hospitals mentioned above and used to describe their disease trajectories over time using the WHO ordinal scale (Ahern et al., 2022 [↗](#)) (**Figure 1B** [↗](#)). In contrast to mild patients, all four severe patients experienced symptoms such as fever, cough, headache, chills, diarrhea, myalgia, and dyspnea (**Table 1** [↗](#)). These patients received mechanical ventilation on sampling D0. In general, symptoms from severe and mild patients diminished gradually up to D28 after recruitment, with the exception of one mild and another severe patient who still experienced mild symptoms.

Mild and severely ill patients displayed different transcriptional programs at the beginning of disease onset. To determine the gene expression profiles of each patient over the time of disease progression, we developed an RNA-seq approach that takes advantage of the longitudinal sampling scheme. By using all expressed genes, we performed a Principal Component Analysis to unbiasedly compare the transcriptional signatures of each patient and two healthy donors. All peripheral blood samples from severe patients on D0 (represented by red circles) were widely dispersed over the left of Principal Component 1 (PC1) (**Figure 1C** [↗](#)), in contrast with mild patients on D0 (represented by blue circles), suggesting that both groups of patients displayed different transcriptional programs at the beginning of the disease. The transcriptional profiles of severely ill patients changed during the recovery phase to be consistent with that observed in mildly ill patients. Gradually, along with disease progression and medical treatments, samples from severely ill patients shifted to the right of PC1 (D7 and D28). Interestingly, on D28, when the majority of patients had recovered, samples from severely ill patients were pooled together with those mild patients who had already recovered. These observations indicated that despite the transcriptional profiles being closer to that of mild patients at D28 as compared to D0, severely ill patients still exhibited higher variability between themselves and controls (**Figure 1C** [↗](#)). In contrast, every mild COVID-19 patient was separated from the severe group on D0 and D7. Notably, only one mild COVID-19 patient (M1) clustered with severe patients at D0. This donor showed a broader set of symptoms over time when compared with the rest of mild patients (**Figure 1B** [↗](#)). This evidence indicated a dissimilar transcriptional response in that specific patient at the onset of disease. Over time, and after medical treatments, the transcriptional program of this patient shifted to be consistent with the other mild patients (**Figure 1C** [↗](#)).

The timing of COVID-19 related gene expression differed between mild and severely ill patients. We focused on the temporal variation of gene expression to identify differentially expressed genes associated with COVID-19 progression. We found statistically-significant differences in the timing of differential gene expression between mild and severely ill individuals (**Figure 1D** [↗](#) and figure supplement 1). We observed that severe patients displayed a transcriptional response completely different from that of mild patients at the sequential time points of D0 and D7 (**Figure 1D** [↗](#)). Previous longitudinal studies identified molecular markers associated with severe COVID-19 (Bernardes et al., 2020 [↗](#); Notarbartolo et al., 2021 [↗](#); Zheng et al., 2020 [↗](#)). We detected these same molecular markers in our severely ill cohort (**Figure 1D** [↗](#)). The expression profiles of those genes varied significantly between mild and severe patients. For instance, the expression of MMP9

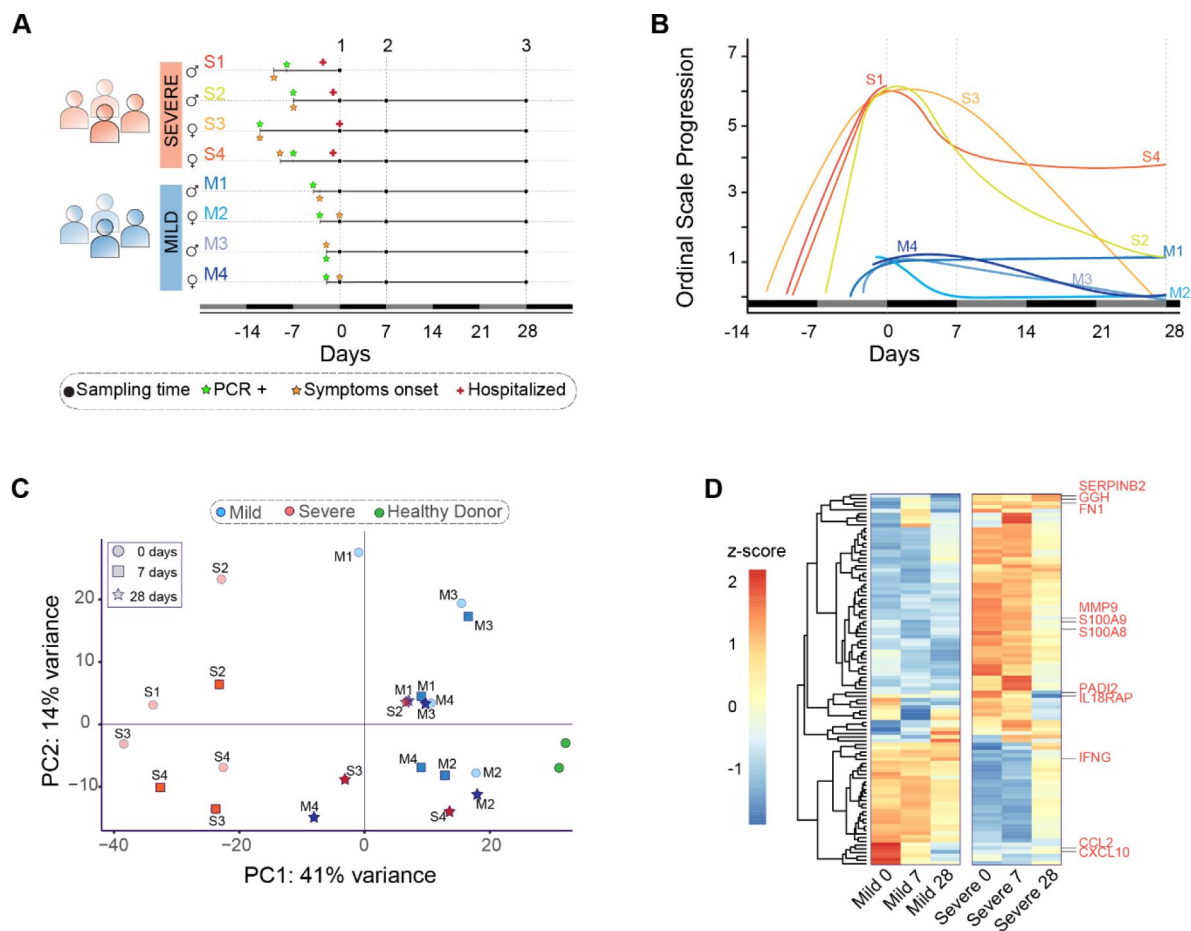


Figure 1.

Clinical profile and gene expression patterns for mild and severe COVID-19 patients during 28 days. **A.** Longitudinal sampling schedule for severe and mild COVID-19 donors of peripheral blood (22 samples in total from 8 donors). Three sampling times (black dots) are displayed with respect to the recruitment day (D0). In addition, panel A shows the diagnosis day with positive PCR (green star), symptoms onset day (orange star), and hospitalization day (red cross). **B.** The COVID-19 progression according to the WHO ordinal scale describes the temporal disease severity for each donor. Severe patients (S1-S4) are displayed in lines with colors scaling from green to red, while mild patients (M1-M4) are shown in blue line colors. The X-axis represents the days relative to the recruitment day. The vertical dot lines indicate the three sampling times used in this study (days 0, 7, and 28). **C.** Principal Component Analysis plot based on gene expression profiles for mild (blue) and severe (red) COVID-19 patients and grouped by their sampling time (0, 7, and 28 days after recruitment). In addition, peripheral blood samples from two healthy donors are shown in green. PC = Principal Component. **D.** Heatmap of the 100 most significant differentially expressed genes related to the COVID-19 disease progression. At the bottom, each column corresponds to the sampling points (0, 7, and 28 days since recruitment) of mild and severe patients. Genes are displayed as horizontal rows and are clustered by the similarity of expression profiles, represented by the dendrogram to the left of the heatmap. Red indicates higher expression, while blue means lower expression represented by the z-score of normalized read counts. Some COVID-19 severity-associated genes previously reported are indicated to the right of the heatmap in red.

CLINICAL CHARACTERISTICS	MILD	SEVERE	ALL
Sex, n (female/male)	(2/2)	(2/2)	(4/4)
Total, n	4	4	8
Median age, years $\pm$ SD	39.0 $\pm$ 3.9	46.7 $\pm$ 8	42.9 $\pm$ 7
Days from onset of symptoms to recruitment, median $\pm$ SD	1.2 $\pm$ 1.3	10.0 $\pm$ 1.8	5 $\pm$ 4.7
Days from COVID-19 diagnosis to recruitment, median $\pm$ SD	3.0 $\pm$ 0.7	8.5 $\pm$ 2.1	6 $\pm$ 3.4
Symptoms, n (%)			
Fever	1 (25%)	3 (75%)	4 (50%)
Chills	1 (25%)	2 (50%)	3 (37.5%)
Fever feeling	1 (25%)	2 (50%)	3 (37.5%)
Odynophagia	2 (50%)	1 (25%)	3 (37.5%)
Cough	1 (25%)	4 (100%)	5 (62.5%)
Expectoration	-	-	-
Dyspnoea	-	4 (100%)	4 (50%)
Thoracic pain	1 (25%)	1 (25%)	2 (25%)
Diarrhea	1 (25%)	2 (50%)	3 (37.5%)
Anosmia	1 (25%)	-	1 (12.5%)
Ageusia	1 (25%)	-	1 (12.5%)
Myalgia	1 (25%)	2 (50%)	3 (37.5%)
Headache	2 (50%)	1 (25%)	3 (37.5%)
Treatment			
Hospitalization, n (%)	-	4 (100%)	4 (50%)
Intubation, n (%)	-	4 (100%)	4 (50%)
Mechanical ventilation, n (%)	-	4 (100%)	4 (50%)
Samples, n	12	10	22

Table 1.

Clinical characteristics of the cohort

metalloproteinase (Zheng et al., 2020 [↗](#)), S100A8/A9 alarmins (Bernardes et al., 2020 [↗](#)), PADI2 (Notarbartolo et al., 2021 [↗](#)) and IL18Rap peptidyl-arginine deiminases (Masood et al., 2021 [↗](#); Schultze & Aschenbrenner, 2021 [↗](#)) were higher in severe patients on D0 than mild or control patients (**Figure 1D** [↗](#)). In addition, we found that *IFNG*, *CCL2*, and *CXCL10* cytokines, which were previously described as molecular markers in severely ill patients (Sette & Crotty, 2021 [↗](#); Vabret et al., 2020 [↗](#)), displayed low expression in our severe COVID-19 patients in comparison with mildly ill patients during the progression of disease (**Figure 1D** [↗](#)).

The immune response of mild and severe patients is activated differentially during the acute phase of the COVID-19 infection

Most of the variations observed in the gene expression profiles of mild and severely ill patients occurred during the acute phase of disease. We performed pairwise gene expression comparisons between mild and severe patients and found differentially expressed genes (DEGs) mainly on D0 and D7. On D0, we found a total of 812 DEGs including 298 up-regulated and 514 down-regulated genes (**Figure 1** [↗](#) - figure supplement 2). On D7, the number of DEGs was similar to D0, with 319 genes showing higher expression and 563 genes with lower expression. We found no differential gene expression between mild and severe patients at D28, supporting the interpretation that most imbalances in the gene expression profiles in the PBMCs of severely ill patients leveled out by D28 (**Figure 1** [↗](#) - supplement 2).

Functional pathways involved with humoral immunity were enriched in severely ill patients during the acute phase compared to pathways involved with cell-mediated immunity in mild patients. The above results provided only a course overview of the transcriptional responses during COVID-19 progression. We expanded our focus to detect molecular mechanisms and pathways involved in the immune responses of all patients by linking functional pathways to differentially expressed genes (DEGs) detected between severely ill, mildly ill and control patients. We used a 2-fold change in gene expression level as a threshold to identify DEGs between mild and severe patients on D0 and D7. We found upregulated expression for genes involved in biological processes that included the T receptor signaling pathway, positive regulation of T cell activation, and regulation of leukocyte cell adhesion in mild COVID-19 patients at D0 (**Figure 2A** [↗](#)). We observed genes involved with immunoglobulin production, antigen receptor-mediated signaling pathway, and adaptive immune response were up regulated at D7 (**Figure 2B** [↗](#)). In contrast, we observed enrichment of gene expression in pathways involved with complement activation, humoral immune response mediated by circulated immunoglobulin, and defense response to bacterium on D0 in severe COVID-19 patients. Furthermore, DEGs in functional pathways mediating hydrogen peroxide metabolic processes, cellular oxidant detoxification and reactive oxygen species were enriched on D7 of infection in this group (**Figures 2A** [↗](#) and **2B** [↗](#)). Biological pathways consistent with a robust lymphocyte cellular immune response were enriched on D0 in mild patients. This functional profile is distinctly different to the antibody / complement-dependent humoral immune responses observed in severely ill individuals at the same time point (**Figure 2A** [↗](#)). Nonetheless, differential expression of genes associated with immunoglobulin function were mainly enriched in mild patients at D7 (**Figure 2B** [↗](#)), while severe patients showed enrichment for genes related to inflammation, reactive oxygen species (ROS), and responses against bacteria at that time of infection (**Figure 2B** [↗](#)).

In addition to enriched biological processes, we also focused on KEGG pathway enrichment among DEGs at D0 and D7 after COVID-19 infection. On D0, mild and severe patients showed considerable differences in terms of the innate response, with the Natural Killer-mediated cytotoxicity pathway enriched in mild-infected patients, while neutrophil extracellular trap formation was enriched in severe ones (**Figure 2C** [↗](#)). Furthermore, DEGs associated with the antigen processing and presentation pathways are enriched in mild COVID-19 patients, in contrast with the enrichment of complement and coagulation cascade pathways in severely ill patients (**Figure 2C** [↗](#)). On the other hand, on D7 of the COVID-19 infections, Natural Killer cell mediated cytotoxicity is one of the main

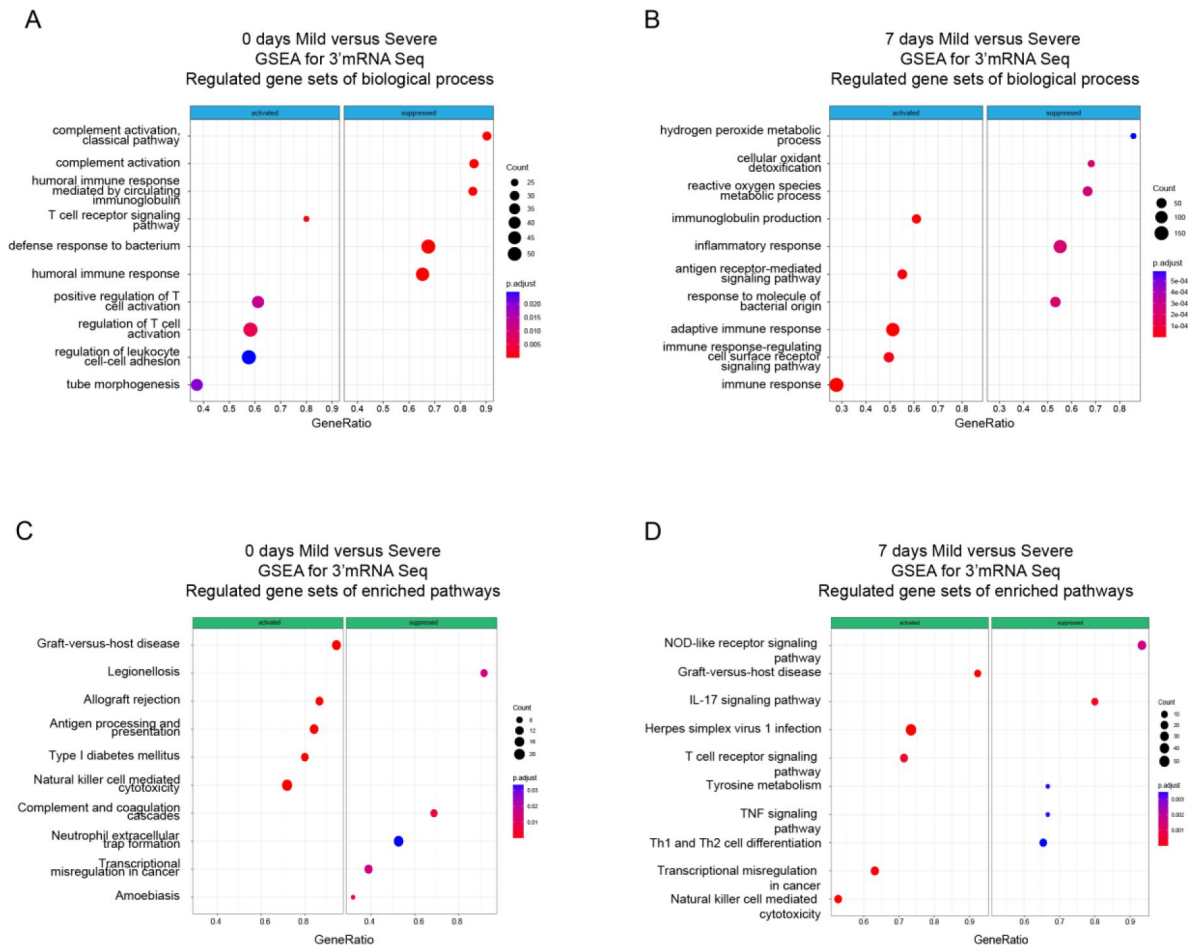


Figure 2.

Gene Set Enrichment Analysis (GSEA) shows biological processes and enriched pathways associated to innate and adaptive immune response after SARS-CoV-2 infection. (**A – B**) Bubble plots show the biological processes (BP) of enriched genes for DEGs between mild and severe for D0 and D7 after recruitment, respectively. (**C – D**) Bubble plots show the KEGG enriched pathways of enriched genes for DEGs between mild and severe for D0 and D7, respectively. The *Count* (black circles) represents the number of genes included on each set. *Generatio* is the ratio between numbers of genes found in the set and total genes of set. The scale-color bar indicates the p-adjustment of each BP or KEGG pathway.

enriched pathways in the mild-infected group, whereas IL-17 signaling is the most significant pathway in the severe-infected group (**Figure 2D**). This finding is remarkable because besides COVID19 IL-17 is affiliated with other clinical pathophysiology in which a dysregulation between innate and adaptive immune responses such as myocarditis and lupus (S. Y. Lee et al., 2019; Rangachari et al., 2006; Sadeghi et al., 2021).

Taken together, we show that there are distinct transcriptional responses along the COVID-19 progression, which suggest that immune responses to SARS-CoV-2 infection occur differently in individuals; thus, there might exist a differential imprinting associated with the severity of the COVID-19 infection.

Higher expression of NK cell hub-genes is a core event of acute phase that distinguishes mild from severe symptoms in COVID-19 individuals

Given that our findings pointed out changes in the immune response after SARS-CoV-2 infection of the patients cataloged as mild and severely ill, we decided to uncover molecular pathways that might be responsible of the differences observed between patient groups during COVID-19 progression. To do so, we first identified genes that were differentially expressed between severity groups, and second, we chose only those that also showed changes in their trajectories across sampling times. In doing so, we found 828 genes that exhibited temporal differences in expression level during disease progression. Then using the Enrichr platform, we discovered additional biological processes and KEGG pathways that were differentially enriched during the COVID-19 progression in mild and severe patients (**Figure 3**). For instance, mild-infected patients exhibited expression of genes involved in kinase activity, enzyme-linked receptor activity, and apoptotic process not only at D0 (acute phase) but also at D7 (middle phase) (**Figures 3A** and **3C**). In contrast, severely ill patients exhibited high level expression of genes involved in neutrophil activity. This observation was the most notorious outcome elicited by SARS-CoV-2 during acute COVID-19 in this group (**Figures 3B**, 3D, and 3F). We observed that Natural Killer cell cytotoxicity was the most enriched pathway among the temporal and differential expressed genes in mildly ill patients during the acute phase (**Figures 3B** and **3C**). Among these enriched genes, we found abundant membrane receptor genes that included *KLRC1*, *KLRC3*, *KLRD1*, *KIR3DL2*, *NCR3*, as well as other intra- and extra-cellular effectors that included *SH2D1A*, *PRF1*, *GZMB*, *FASLG*, *ZAP70*, *IFNG*, *CD247*, and *LAT*. Furthermore, the *ZAP70*, *CD4*, *IFNG*, *IL2RB*, *STAT4*, *CD247*, *DLL1*, *LAT*, and *IL12RB2* genes were enriched in COVID-19 mild patients during the acute phase (**Figure 3** - table supplement 1 and 2). This data indicated that the Th1/Th2 cell differentiation pathway was robust and active during this phase and likely played an important role in the effective adaptation to dynamic events during the progression of the infection that protected mildly ill patients from experiencing severe symptoms. Interestingly, metabolic pathways involved with hematopoietic cell lineages were enriched in severely ill patients at this matched moment in time with the mildly ill patients (**Figure 3B** and **3D**). Collectively, these observations indicate that coordination between humoral and cell mediated immunity were more tightly regulated in mildly ill patients than in severely ill patients. To confirm the importance of the differentially enriched pathways between mild and severe COVID-19 patients, we focused on analyzing the context of gene-gene interactions (**Figure 4A** and **Figure 4** - figure supplement 1) and changes in their quantitative expression levels overtime graphed as a heatmap (**Figure 4B**). The genes displayed in this KEGG pathway graph represent the up-regulated genes (red boxes) in mild patients and their interactions involved in NK cell-mediated cytotoxicity (**Figure 4A**). Interestingly, all these genes showed overtime trajectories with high levels on days 0 and 7 in mild patients. These gene expression levels became roughly equivalent by D28 in both the mild and severe groups (**Figure 4B**). Complementing these observations, we constructed a protein-protein interaction (PPI) network using only upregulated genes during the early phase (days 0 and 7), followed by a clustering process that detected proteins with more significant interactions among the selected genes (**Figure 4C**). Notably, we detected *KLRD1*, *CD247*, and *IFNG* as central

nodes of protein-protein interaction networks. This finding makes sense because these proteins exhibit numerous interactions with other proteins involved in activating or inhibiting NK cell cytotoxicity (e.g., KLRC1, KLRC3, and KIR3DL2), as well as Th1/Th2 cell differentiation (CD4) and cytokine-cytokine receptor interaction (IL5RA, IL2RB). In **figure 4D**, we show the comparative trajectories of these node-genes between both groups of severity. Interestingly, we found a convergence of KLRD1 and CD247 genes on D28, while IFNG remained differentially expressed between patient groups.

Once we identified the trajectories of NK cell hub-genes participating in COVID-19 progression, we asked whether there were any DEGs (adj. $p \leq 0.05$ and $\log_2$ -fold change ≥ 2.0) obtained from a pairs-wise comparison of mildly and severely ill patients at days 0 and 7 that would have been left out from the longitudinal analysis. Given that the number of DEGs at each time point is higher when compared to the list of genes exhibiting differential trajectories, we performed a GO and pathways analysis with the new set list of genes (**Figure 4** - figure supplement 2). The main result showed that Natural Killer cell mediated cytotoxicity was predominant on D7. This finding reinforced the interpretation that there is a dysregulation of innate immunity, as previously suggested in severe patients (Paludan & Mogensen, 2022), with an over-representation of neutrophil activation. The results shown in **Figure 4** - figure supplement 3 and 4 summarize the pathway and PPI network analysis for these genes on D0 and D7, respectively, and show the predominant enrichment of NK genes. Taken together, these data are consistent with an active and regulated innate NK cytotoxic immune response mounted during the acute phase of infection in mild COVID-19 patients. This observation contrasts with the humoral- and neutrophil-biased response observed in severely ill individuals.

Previous comparisons were done with the assumption that both cohorts were at the peak of their symptoms on D0. However, taking into account the delta at the days of symptoms onset, we also analyzed the pairwise comparison for D7 in mild patients, but this time comparing it to D0 in severe patients. The **Figure 4** - figure supplement 5 highlights GO terms related to Natural Killer cell response and enriched pathways for NK cell cytotoxicity in mild patients. This result supports the idea that the transcriptional program differs between mild and severe individuals with the outstanding contribution of NK cells in mild patients. Otherwise, severe patients exhibit an inflammatory response reflected by the complement and coagulation pathways, which is according to our previous analysis.

Gene co-expression identifies NK hub genes linked to the innate and adaptive immune response of mild COVID-19 patients

We identified genes that were coordinately expressed during COVID-19. We developed a weighted gene correlation network (WGCN) to simultaneously analyze all peripheral blood samples collected from patients during the longitudinal protocol and those from healthy donors to identify genes with coordinated expression. By using a differential co-expression approach, we identified ten modules of co-expressed genes (**Figure 5A**). We then used these networks to correlate each module with available clinical information of the patients by calculating the module significance (MS) for each module-trait correlation. Not surprisingly, we found that most module eigen genes grouped according to the degree of COVID-19 infection (i.e., mild or severe patients) (**Figure 5B**). Among co-expressed gene modules, we focused on three modules that contained the largest number of genes. These modules correspond to blue (704 genes), brown (508 genes) and turquoise (712 genes). The blue and brown modules, which are correlated positively with mild patients (**Figure 5B**), were enriched with genes related to T-cell activation and platelet function, respectively (**Figure 5C**). In contrast, the turquoise module, which was correlated positively with severe COVID-19 patients (**Figure 5B**), was enriched with genes related to neutrophil activation and inflammatory responses (**Figure 5C**). As shown thus far, the Th1/Th2 cell differentiation pathway was relevant in the immune response of mild COVID-19 patients, and because the blue module is enriched in lymphocyte-based immune response genes, we performed

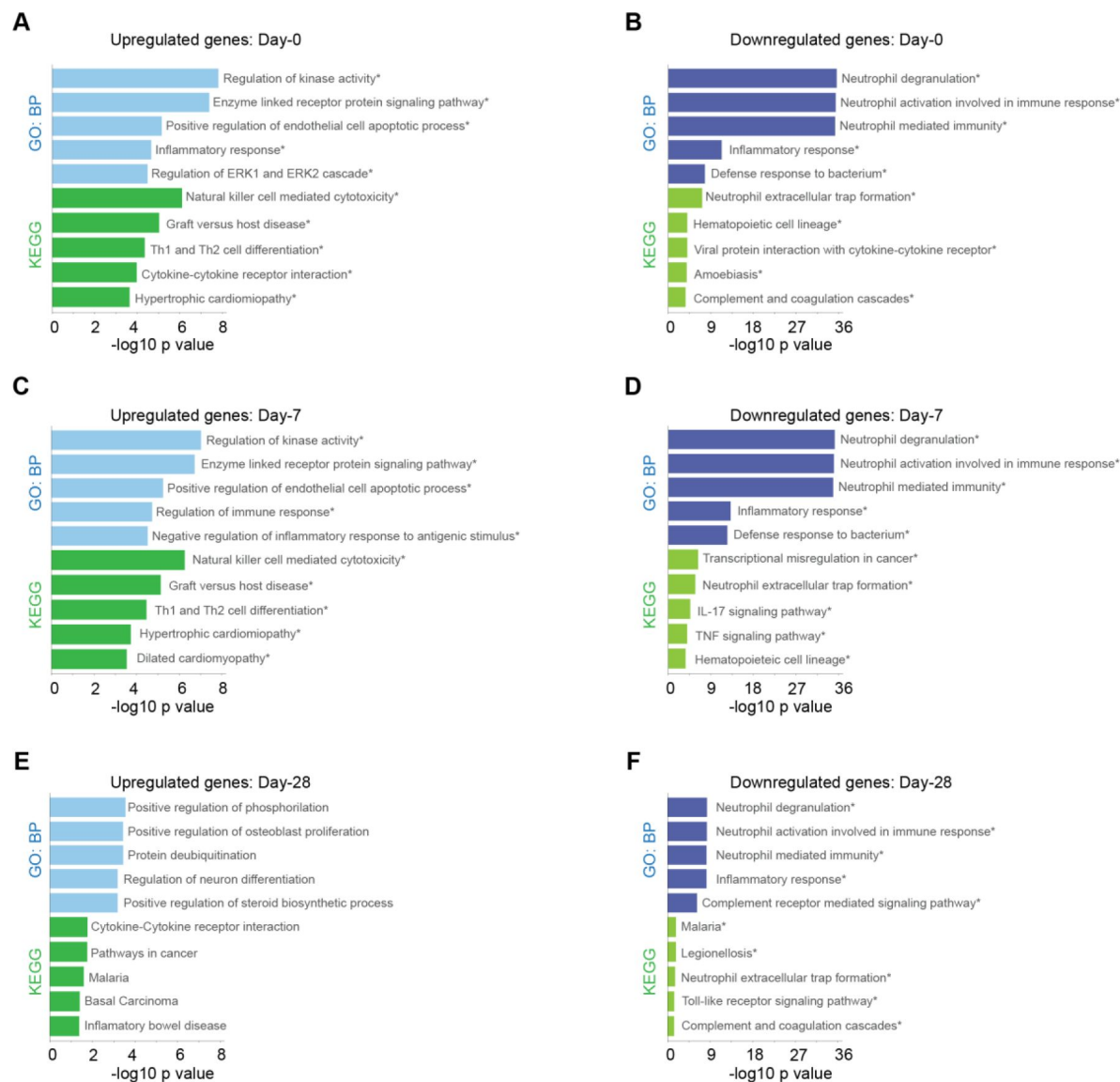


Figure 3.

Biological processes and KEGG pathway for genes with differential expression levels over time in mild versus severe patients. Bar plots show Gene ontology analysis for biological processes in light-blue/dark-blue and KEGG pathway in light-green/dark-green, using upregulated genes (**A - C - E**) and downregulated genes (**B - D - F**). The size of each bar is according to its $-\log_{10}$ p-value, and name pathways with asterisk indicates a q-value ≤ 0.05 .

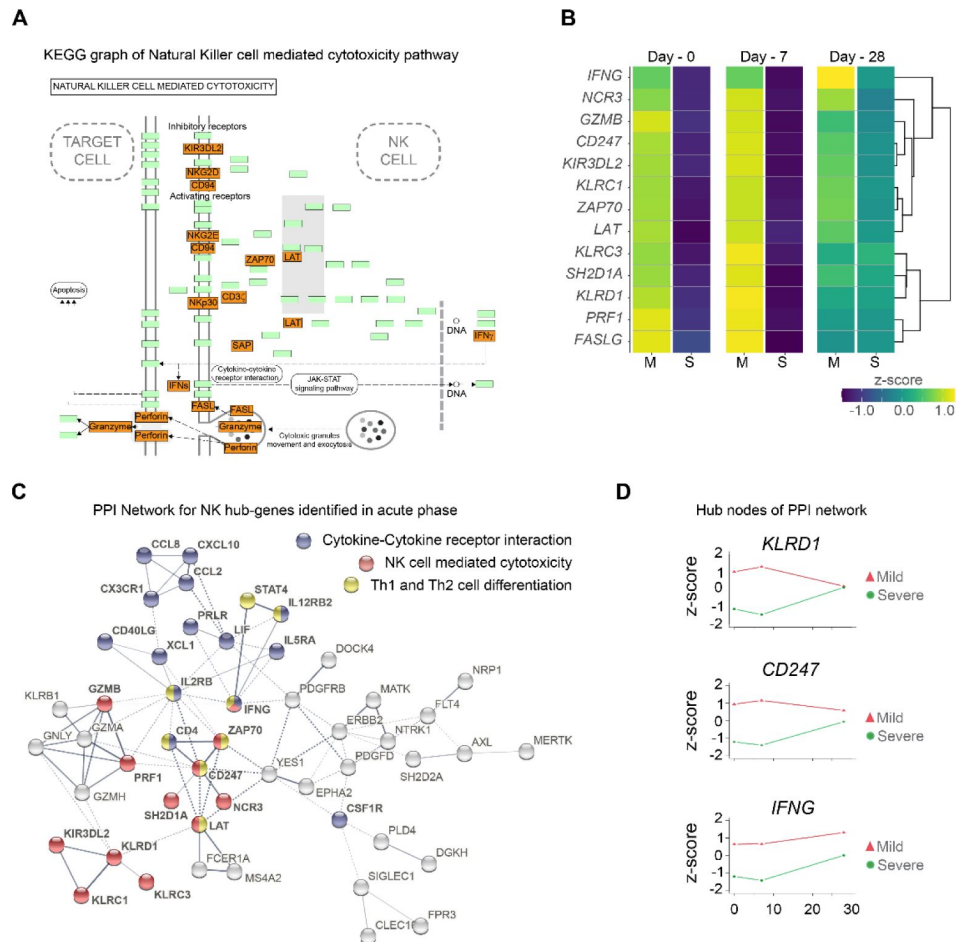


Figure 4.

Gene function network for Natural Killer cell hub-genes with differential expression levels between mild and severe patients during acute phase of COVID-19. **(A)** KEGG pathway of Natural Killer cell mediated cytotoxicity represents the set NK cell hub-genes upregulated (red-boxes) in mild versus severe patients. The green boxes correspond to genes without differential expression. **(B)** Heatmap shows the differential expression levels of NK cell hub-genes over time (D0, D7, and D28 after recruitment) separated by mild and severe groups. The expression levels are represented by the z-score of normalized counts. Dendrogram shows the hierarchical clustering of genes. **(C)** Protein-protein interaction (PPI) network for upregulated genes during the acute phase in mild patients. The network corresponds to the principal clusters with more interaction between proteins and highlight the three most represented pathways: Cytokine-cytokine receptor interaction (blue); NK cell mediated cytotoxicity (red); and Th1 and Th2 cell differentiation (yellow). **(D)** Time course expression levels for the main protein nodes identified in PPI network during the acute phase of COVID-19. The trajectories of these genes are graphed as days after recruitment (D0, D7, and D28) for mild (red triangle) and severe (green circle) groups and their enrichment is represented by the z-score of normalized counts.

a gene-gene network analysis to determine how the genes from this module might have worked in the context of an adaptive immune response. In this analysis, we found genes belonging to the NK cell-mediated cytotoxicity pathway grouped together with the cytokine-cytokine receptor interaction and Th1/Th2 cell differentiation pathways (**Figure 5D** [↗](#)). Furthermore, these genes were previously identified as differentially expressed in the NK cytolytic pathway, like *KLRD1*, *KLRC3*, and *KLRC1* receptors, as well as *FASLG*, *SH2DB1A/B*, and *LAT*. All of this evidence is consistent with the interpretation that highly interconnected genes from blue module had a functional significance in limiting the progression COVID-19 in mild patients. In the other modules, the brown network (**Figure 5E** [↗](#)) depicts enriched pathways for platelet activation, extracellular matrix-receptor interaction, hematopoietic cell lineage, gap junction, and complement and coagulation cascades which complemented with GO terms is an important focus of interest in COVID19 (S.B et al., 2024). Furthermore, the enriched pathways for the turquoise module include leishmaniasis, malaria, osteoclast differentiation, and nitrogen metabolism (**Figure 5F** [↗](#)), some of which are implicated in a neutrophil response as indicated by the GO terms (Babatunde & Adenuga, 2022 [↗](#); Passelli, Billion, & Tacchini-Cottier, 2021 [↗](#)).

The remaining seven modules were analyzed for Gene Ontology (**Figure 5** [↗](#) - figure supplement 1) which reveals different aspects related to the immunological response to viral infection. In addition, we searched for enriched pathways and displayed the results in the form of a network graph (**Figure 5** [↗](#) - figure supplement 2) that can be further investigated in future studies.

Discussion

We systematically analyzed transcriptomic features of PBMCs from COVID-19 patients with mild and severe symptoms at three sequential time-points (D0, D7, and D28) during the peak of the symptoms. Our longitudinal analysis revealed key temporal features of immune responses that distinguished mild from severe patients during acute disease. We observed a prominent role of Natural Killer (NK) cell mediated cytotoxicity function pathways during COVID-19 progression. These pathways include genes such as *KLRC1*, *KLRC3*, *KLRD1*, *KIR3DL2*, and *NCR3* receptors, as well as other effectors like *SH2D1A*, *PRF1*, *GZMB*, *FASLG*, *ZAP70*, *IFNG*, *CD247*, and *LAT*. Most, if not all, of these genes are implicated in regulatory processes of cytotoxicity and attraction of NK cells as part of the viral infection control mechanism (Björkström, Strunz, & Ljunggren, 2022 [↗](#)). Antiviral NK cell cytotoxicity depends on a steady state for survival, basal turn-over and their function maintenance, which are monitored by several checkpoints (Björkström et al., 2022 [↗](#); Masselli et al., 2020 [↗](#); Vivier, Tomasello, Baratin, Walzer, & Ugolini, 2008 [↗](#)). We found a dynamic transcriptomic profile of a NK cell gene-hub characterized by higher gene expression levels in individuals with mild disease compared with those with severe symptoms across 0 and 7 days. However, expression levels of these NK cell gene-hubs became more similar between mild and severe patients on D28. In contrast to this orchestrated transcriptional response of dominant NK cells activities, we found an up-regulated gene signature consistent with dominant neutrophil activities in our severe cohort even after recovery. This finding was previously observed with the concomitant increase of IgG production and complement activation at the earliest phase of disease (J. Wang et al., 2020 [↗](#); B. Zhang et al., 2020 [↗](#); Zuo et al., 2020 [↗](#)), (**Figure 2** [↗](#)).

In our NK cell gene hub, we recognized activating (*KLRC3*, *NCR3*), and inhibitory (*KLRC1*, *KIR3DL2*) genes of cytotoxicity, as well as regulatory and effector proteins (*KLRD1*, *GZMB*, and *PRF1*). All these genes could be participating in the balancing of a well-coordinated NK cell activity profile. In this sense, the *KLRD1* gene, which encodes the CD94 protein, stands out as an important node interconnecting proteins networks. This node regulates activating (NKG2E from *KLRC3* gene) and inhibitory functions (NKG2A from *KLRC1* gene), and thus modulates NK cell cytotoxicity (Borrego, Masilamani, Marusina, Tang, & Coligan, 2006 [↗](#)). Supporting this role, a previous study demonstrated the importance of CD94:NKG2E heterodimeric receptor in response to the lethal mousepox virus (Fang et al., 2011 [↗](#)). This node may be relevant for an efficient response against

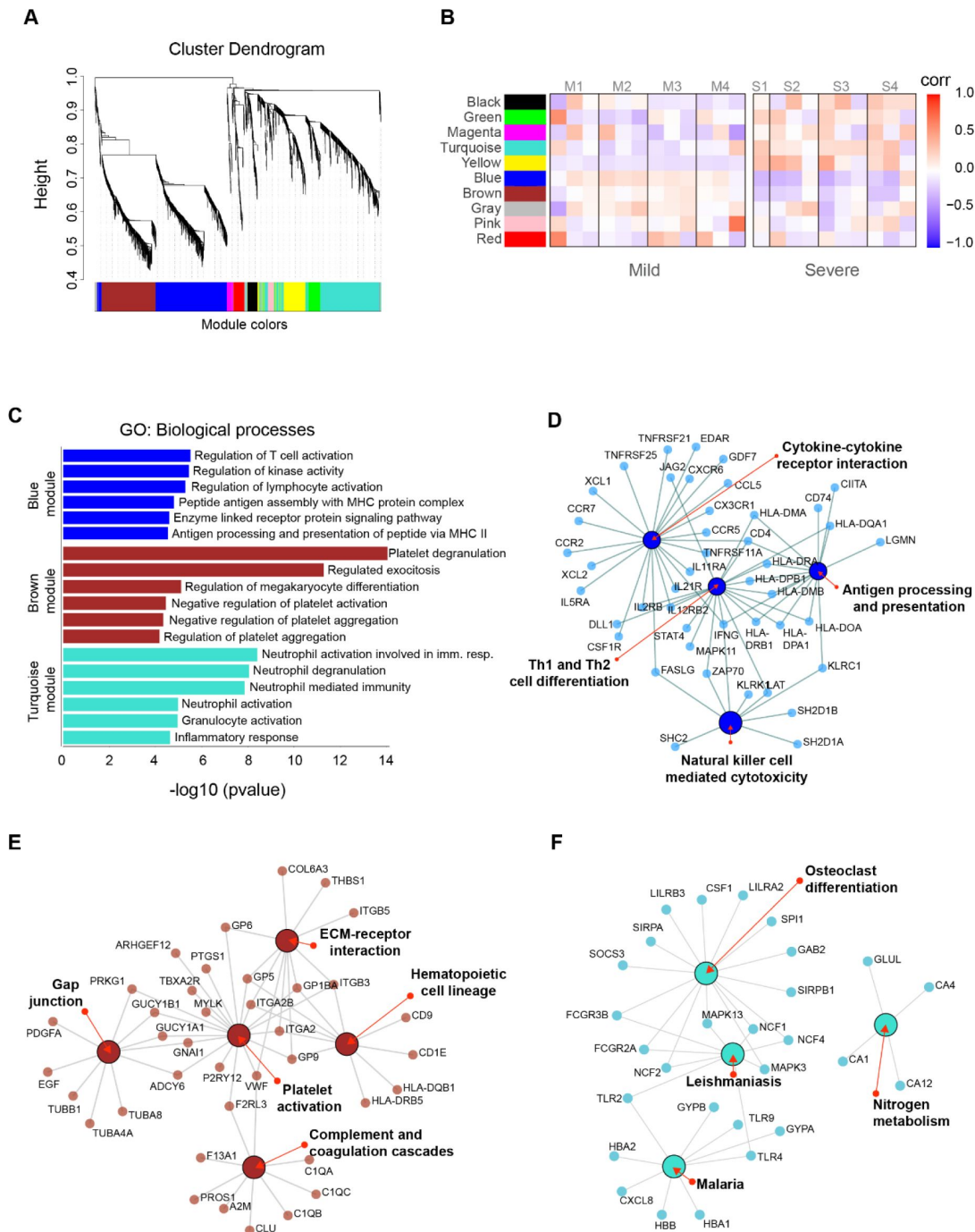


Figure 5.

Gene co-expression network analysis among the longitudinal transcriptomic profiling. **(A)** Gene hierarchical clustering dendrogram of ten detected modules based on Topological Overlap Matrices (TOM) measure. The branches and color bands represent the assigned module. **(B)** Module-trait relationships (MTRs) between detected modules and disease severity of COVID-19. MTRs are obtained by calculating Pearson correlation between the traits and the module eigengenes. The red and blue colors indicate strong positive or negative correlations, respectively. Rows represent module eigengene (ME) and columns indicate the disease severity of COVID-19, where correlations from each patient are shown from left to right according to their sampling time (D0, D7, and D28). **(C)** GO enrichment analysis of genes in the blue, brown, and turquoise modules. The color in the bar graphs refers to the module eigengene (ME). Enrichment results are sorted by $-\log_{10}(\text{p-value})$ (higher on top) of each biological process GO term. **(D-E-F)** Network visualization of enriched pathways based on co-expression genes from blue, brown, and turquoise modules, respectively.

SARS-CoV-2 infection given the high conservation of receptors and ligands between the human and mouse pathways (Borrego et al., 2006 [DOI](#)). Wauters et al. found that mild COVID-19 patients displayed an interaction of CD94:NKG2E/HLA-F between their T cells and neutrophils in bronchoalveolar lavage samples (Wauters et al., 2021 [DOI](#)). On the other hand, NKG2A is an important inhibitory receptor that interacts with CD94 and together regulate NK cell functions (Borrego, Masilamani, Kabat, Sanni, & Coligan, 2005 [DOI](#); N. Lee et al., 1998 [DOI](#)). Our analysis showed a higher expression of NKG2A in mild than severe patients during the acute phase. Regarding the same receptor, previous research showed that NKG2A was more highly expressed in lymphocytes and NK cells during infection compared with healthy controls (Zheng et al., 2020 [DOI](#)). In parallel, Zheng et al. showed a decrease of expression of NKG2A in recovering patients along with an increase of NK cell number. Collectively, this evidence supports the conclusion that CD94, and its partners, play important roles in regulating both activating and inhibitory checkpoints related to NK cell cytolytic functions. Furthermore, it substantiates the relevance of innate NK cell immune responses in combating SARS-CoV-2, and likely other coronaviruses. This pathway might also be a prominent player in controlling other infectious respiratory virus infections to promote a mild presentation of disease.

Other genes located in the NK cell hub included membrane proteins such as *SH2D1A*, *LAT*, *CD247*, *FASLG*, the enzyme *ZAP70*, and the cytokine *IFNG*. Remarkably, genes encoding *IFNG* and *CD247* were also identified as important nodes within the protein-protein interaction network during the acute phase. Considering the interactions of these nodes with proteins involved in cytokine-cytokine receptor interactions and Th1/Th2 cell differentiation pathways, it is possible that they coordinately regulated these immune responses with NK cell cytolytic functions. In this context, cytokine-cytokine receptor interaction and Th1/Th2 cell differentiation were well-represented pathways in mild patients during the acute phase highlighting that both innate and adaptive immune responses were active and effective in these patients. Particularly, CD247 (CD3 ζ) protein is part of the T-cell antigen receptor (TCR) complex, whose low expression levels have been related to chronic inflammation and decreased T cell activity (Y. Li et al., 2021 [DOI](#)). In the same line, IFNG protein is a critical player between innate and adaptive immunity after viral infection (Kang, Brown, & Hwang, 2018 [DOI](#)). Giving support to this connection between innate and adaptive immune responses, it would be expected that adaptive CD8⁺ T cell cytolytic functions would also be enriched in mild patients due to its important role controlling viral infections (Prager & Watzl, 2019 [DOI](#); Uzhachenko & Shanker, 2019 [DOI](#)). Interestingly, GO/pathway-based analyses did not detect these functions as a differential player in clinical COVID-19 progression, despite the fact that some genes are shared with NK cytotoxic gene hub (Uzhachenko & Shanker, 2019 [DOI](#)). This observation suggested that Th1/Th2 cell differentiation may be more essential for a successful adaptive response against SARS-CoV-2 than CD8⁺ T cell cytolytic function in mild patients, at least during the early phase of COVID-19. If this interpretation is credible, then cell-mediated cytolytic activities should rely on the well-regulated activity of innate NK cell subset as a primary immune response. Taking into account all these data, it is reasonable to interpret that an early fate-compromise towards NK cell activity instead of a Neutrophil effector activity may have had an important effect on subsequent processes regulating adaptive immunity. This model favors a robust integration of innate and adaptive immune response during an effective control of COVID-19.

Until now, we have discussed relevant genes involved in immune pathways enriched in mild or severe COVID-19 progression. However, we also decided to look for genes exhibiting coordinated gene expression patterns across all our samples. We found that one module (blue module), which has a strong positive correlation with mild patients, included genes involved with metabolic pathways regulating T cell activation, kinase activity, and antigen presentation. In contrast, the turquoise module, which exhibited a strong positive correlation with severely ill individuals, contained genes associated with neutrophil-related biological processes. This finding was indicative of an opposed early fate of innate immune responses between mild and severe COVID-19 cases. Neutrophil long-term differential enrichment seen across severe cases could be related to other repercussions of SARS-CoV-2 infection, like neutrophil-induced platelet aggregation (Ievtic &

Nazy, 2022 [\[4\]](#)). Consistent with this interpretation, dysfunction of platelets has been associated with abnormal clot formation in severe COVID-19 cases (Litvinov et al., 2021 [\[4\]](#)). In this sense, our results show that the brown module, which is negatively correlated with severe patients, displays biological processes linked to platelet degranulation activity and negative regulation of aggregation. Hence, these results are consistent with a platelet dysfunction pathology linked to severely ill patients.

We performed a pathway enrichment analysis to understand how the positive correlation of genes in the blue co-expression module related to immune response functions in mildly ill patients. Not surprisingly, the main pathways enriched included Th1/Th2 cell differentiation, cytokine-cytokine receptor interaction, antigen processing, and NK cell-mediated cytotoxicity pathways. Albeit the co-expression analysis included all samples, regardless of the severity of the disease or the longitudinal sampling. This effort revealed transcriptional programs of immune response that were consistent with the profiles detected in mild and severe patients reported in our previous investigations. This evidence supports the idea that the transcriptional regulation of cell mediated immunity in mildly ill patients is more robust than that observed in patients with severe clinical progression.

In order to identify differences in transcriptional programs associated with mild or severe outcomes, we carefully compared changes in gene expression during the acute phase. This analysis detected a broader list of genes than those found using the longitudinal analysis alone. This accomplishment was resulted from only considering the differences in gene expression between mild and severe groups, independently of quantitative changes in gene expression over-time. These results consistently showed a NK cell hub of genes being differentially expressed in mild patients. Importantly, we found novel DEGs including *KLRK1*, *KIR2DS4*, and *KLRC2*. The gene *KLRK1* codes for NKG2D protein, an activating receptor with critical importance due its interaction with the major histocompatibility complex-class-I (Zingoni et al., 2018). We found that NKG2D is comparatively over-expressed between days 0 and 7 in the mild group. In line with this finding, Varchetta et al. found an increase of circulating NKG2D(-) NK cells using cell cytometry during the acute phase, which was linked to exhaustion in severe COVID-19 patients. Importantly, their sampling times ranged from hours to days after onset symptoms (Varchetta et al., 2021 [\[4\]](#)). Additionally, the regulatory role of NKG2D in COVID-19 is also supported by Lee et al. where their results show that the viral non-structural protein 1 (Nsp1) of SARS-CoV-2 mediates its immune escape by downregulating NKG2D-Ligands, therefore decreasing NKG2D-dependent NK cytotoxic responsiveness and conferring resistance to infected cells (M. J. Lee et al., 2022 [\[4\]](#)).

Complementary to this scenario of NK cell activating receptors being important in mitigating symptom severity as previously reported (Gardiner, 2008 [\[4\]](#)), we found that the *KIR2DS4* gene, which belongs to the KIR receptors gene-family, was correlated with mild progression of COVID-19. This activating receptor was more highly expressed in mild patients than in severe patients at both 0 and 7 days (4-fold and 6.8-fold, respectively). In this regard, Bernal et al. found that a low expression of *KIR2DS4* was part of a distinctive immunophenotype in peripheral NK cells that was increased in severe COVID-19 individuals (Bernal et al., 2021 [\[4\]](#)). On the contrary, Casado et al. found an enriched KIR2DS4(+) subset of CD56brightCD16neg peripheral NK cells in hospitalized individuals compared to mild patients, indicating a positive correlation with severity (Casado et al., 2022 [\[4\]](#)). However, as their mild patient cohort were recruited at a mean of 60 days after diagnosis, a direct comparison with our data is not precise. Additionally, the absence of severe patients, as well as a lack of longitudinal sampling, were some of the inconsistencies between the study designs of Casado et. al. and ours, which may account for these inconsistent observations.

Another NK cell activating receptor (the NKG2C protein) encoded by the *KLRC2* gene was previously implicated as a COVID-19 marker (Fielding et al., 2022 [\[4\]](#); Maucourant et al., 2020 [\[4\]](#); Vietzen et al., 2021 [\[4\]](#)). Although we did not find a significant difference in the gene expression of *KLRC2* between mild and severe groups (being excluded by our threshold criteria), we found it to

be comparatively lower among severe patients compared to mild patients in both 0 and 7 days (-1.6 fold and -2.7 fold, respectively). We observed a similar trend in the control patients between 0 and 7 days in the acute phase (-1.6 fold and -1.7 fold, respectively). Surprisingly, when we compared the quantitative expression of the NKG2C gene in the severe group to the control group on D28, we discovered an inverted pattern with respect to the acute phase in which NKG2C overpassed the levels of controls (2.1-fold). In this context, Maucourant et al, in a scRNA-Seq study performed with bronchio-alveolar lavage (BAL) from severe COVID-19 patients, showed increased NKG2C levels linked to the adaptive response of NK cells (Maucourant et al., 2020 [DOI](#)). These data additionally reinforce the interpretation that NKG2C is required to mount an effective NK cell response against SARS-CoV-2 infection. Vietzen et al. found that a deletion in the NKG2C gene resulted in a significant correlation with severe COVID-19 (Vietzen et al., 2021 [DOI](#)). This evidence supports the idea that the innate and adaptive immune responses are being differentially modulated in severe COVID-19 than mild patients. Another important observation detected in our comparative analysis included enrichment of the IL-17 pathway in severe patients on D7. Our analysis identified a group of 10 genes (*FOSL1*, *CXCL6*, *CEBPB*, *LCN2*, *TNFAIP3*, *CXCL1*, *CXCL2*, *MMP9*, *S100A9*, and *S100A*) associated with this time point. IL-17 has diverse biological functions that promote protective immunity against many pathogens but also driving inflammatory pathology during autoimmunity. Interleukin-17-driven inflammation is normally controlled by regulatory T cells expressing the anti-inflammatory cytokines IL-10, TGFbeta, and IL-35 (Pacha, Sallman, & Evans, 2020 [DOI](#)). One explanation may be that an imbalance in T cells and cytokine secretions mediated by IL-17 promote an inflammatory phenotype in patients with severe symptoms. Notably, Th17 cells were elevated on D7 in patients with mild symptoms. These Th17 cells can display plasticity in cytokine production *in vivo* and can switch from predominantly producing IL-17 to predominantly producing IFN γ , thereby resembling Th1 cells (Y. K. Lee, Mukasa, Hatton, & Weaver, 2009 [DOI](#)). Sequential activation of STAT1 by IFNG and STAT4 by IL-12 drives optimal expression of T-bet (TBX21), a central transcription factor for Th1 programming (Y. K. Lee et al., 2009 [DOI](#)). Otherwise, the activation of STAT6 by IL-4 upregulates GATA3, which is central to Th2 programming (Y. K. Lee et al., 2009 [DOI](#)). All the genes related to Th1 activity (*IFNG*, *STAT4*, *TBX21*, and *IL-12*) were up-regulated in our cohort of patients exhibiting mild symptoms, which is consistent with this potential regulatory circuit.

Even though the primary criterion for comparing mild with severe was the peak of symptoms in earlier analyses, we also investigate according to the time of the symptoms onset. As a result, we also performed a complement analysis comparing mild-D7 to severe-D0 to identify the enriched pathways that represent this comparison. Notably, the most important GO-terms and enriched pathways in mild patients correspond to Natural Killer cell cytotoxicity, represented by the same set of genes reported in previous comparisons. This lends significant support to the primary findings of our study, which show that a transcriptional program is differentially regulated in mild patients and is focused on innate immune response associated with Natural Killer cell cytotoxicity.

In conclusion, the longitudinal trajectories of gene expression, the differential GO/Pathways, and protein-protein interactions analyses, together with the co-expressed gene-gene correlation network is consistent with the existence of a regulatory transcriptional program linked to an early activation of NK cell cytotoxicity in mild COVID-19 patients. This work establishes the notion that innate immune responses are crucial for the progression of COVID-19 severity, and reinforces the importance of the NK cell cytotoxicity pathway in distinguishing between mild and severe COVID-19 progression. Taken together, these differential responses are complemented by cytokine activities and Th1/Th2 cell differentiation programs indicating a well-regulated crosstalk between innate and adaptive immune responses in the mild COVID-19 progression.

Methods

Patient cohort and Peripheral Blood Mononuclear Cells (PBMCs) sampling

We recruited a total of 8 patients diagnosed to be suffering from COVID-19 who were separated into two groups, one composed of 4 mild outpatients and another, conformed of 4 severe hospitalized individuals. Peripheral venous blood samples were obtained by using the venipuncture technique in Vacutainer K2 EDTA tubes (BD, USA) from each patient three times, including two clinical stages (acute phase and convalescence). PBMC were isolated from each fresh heparinized peripheral blood sample, through density gradient centrifugation on Ficoll-Paque Plus (GE Healthcare Life Sciences, USA) by centrifuging at 1600 rpm for 30 minutes (using minimum acceleration and no deceleration configurations). PBMC-containing fraction was then washed two times with 2 mM EDTA in PBS and stored in RNeasy lysis solution (Qiagen, USA) at -20° C until RNA extraction. The detailed clinical features of all patients and the detailed sampling time are shown in [Figure 1](#) and [Table 1](#). All samples were processed in a qualified BSL-2 laboratory and, according to protocols and approval from Institutional Review Boards, CEC-SSLR Ord N°226 and Ord N°399. Written informed consent was received before the participation of each patient.

RNA extraction, library preparation, and PBMC transcriptome sequencing

Total RNA extraction was performed from PBMC by Diagenode (Belgium). RNA samples were quantified using Qubit™ RNA BR Assay Kit (Thermo Fisher Scientific, USA) and secondly checked for integrity using HS RNA Kit (Agilent, USA) on a Fragment analyzer system (Agilent, USA). The library preparation was performed using NEBnext ultraII Directional Kit and sequencing of the samples was performed on an Illumina NovaSeq 6000 instrument producing 150bp paired-end reads running Control Software 1.7.0.

Identification of differentially expressed genes along COVID-19 progression and between disease severities

Sequencing-quality check was performed using FastQC ([Andrews, 2010](#)), and low-quality reads were trimmed using Trim_Galore! (Krueger) with --clip_R1 3 and --clip_R2 3 options. High-quality reads were aligned to the human reference genome version GRCh38 with STAR v2.6.1a_08-27 ([Zhang et al., 2022](#)). Transcript counts were generated using featureCounts v1.6.3 ([Bastard et al., 2020](#)) with default settings. Differential gene expression analysis was performed in two ways using edgeR package v3.36.0 ([Delorey et al., 2021](#)). Temporally and differentially expressed genes and differentially expressed genes between mild and severe COVID-19 patients at each sampling timepoint were identified using the generalized log-linear model (GLM) option in edgeR. Genes were considered as differentially expressed either with temporal expression differences or disease severity condition using a false discovery rate (FDR; Benjamini–Hochberg), an adjusted *p*-value of <0.05, and an absolute log2 fold change of 2. Transcript counts (normalized using TMM approach) were used to generate heatmaps for visualization of differentially expressed genes using the pheatmap R package. Expression was scaled by row z-scores for visualization, taking into account the mild and severe patients to obtain the mean and standard deviation of each group.

Gene set enrichment analysis (GSEA)

In order to perform GSEA ([Schultze & Aschenbrenner, 2021](#)), after DEGs analysis only values of log2 fold change equal to or greater than 1 were considered. Then the GSEA was performed through ClusterProfiler package (v.3.16) ([Yu, Wang, Han, & He, 2012](#)) in R, we evaluated the

significantly enriched biological process (GO) using gseGO function and pathways from Kyoto encyclopedia of gene and genomes (KEGG) using gseKEGG function.

Enrichment analysis of differentially expressed genes along COVID-19 progression and between mild and severe patients

Gene ontology (GO) and pathways enrichment analyses were performed with both upregulated and downregulated genes using Enrichr platform (Xiong et al., 2020 [\[1\]](#)). Significant GO terms (Biological processes and Molecular functions) and pathways (KEGG and Reactome) were calculated from the adjusted p-value (q-value) using the Benjamini-Hochberg method for correction for multiple hypothesis. Considering a difference of one unit in z-score between the two severity groups, for mild patients we analyzed 365, 359, and 101 genes for days 0, 7, and 28 respectively, while for severe patients we analyzed 369, 414, and 136 genes for day 0, 7, and 28 respectively.

Weighted correlation network analysis (WGCNA)

Based on the assumption that differentially expressed genes may explain transcriptional differences observed between mild and severe COVID-19 patients. Read counts from differentially expressed genes among all samples were selected as a reference set for construction of a weighted gene co-expression networks and modules detection. Co-expressed gene modules were constructed using WGCNA R package v1.71 (Langfelder & Horvath, 2008 [\[2\]](#)) under a signed networks approach because it provides a better understanding of molecular regulatory mechanisms at the systemic level, facilitating better separation of modules in terms of biological performances. To do so, we removed outliers using the adjacency function and a standardized connectivity score of < -2.0 ; then we used the pickSoftThreshold function to identify the soft thresholding power β value, which was subsequently transformed into a Topological Overlap Matrix (TOM). Next, an average linkage hierarchical clustering analysis was performed based on the TOM dissimilarity (1-TOM) and modules were detected through a dynamic hybrid tree cutting algorithm. After the modules were identified, the module eigengene (ME) was summarized by the first principal component of the module expression levels. Module-trait relationships (MTRs) were estimated using Pearson correlation between MEs and disease severity. To evaluate the correlation strength, we calculated the module significance (MS) that is defined as the average absolute gene significance (GS) of all genes involved in a module.

PPI network

All genes with differential trajectories over time (#827 genes) were included to construct a protein-protein interaction network (PPI) by STRING V11.5 (Szklarczyk et al., 2020 [\[3\]](#)). After clusterization, we focused our analysis on the principal cluster (1 of 3) with the highest connectivity between genes and highlighted the genes involved in the most significant pathways according to KEGG.

Acknowledgements

We especially thank all blood donors who participated in this study. COVID-19 South Chile Group: Renato Ocampo, Christian Esveile, Leonila Ferreira, Johanna Cabrera, Geraro Solís, Marcelo Pozas, Equipo de enfermería Servicio de Medicina Interna Hospital Base Osorno, María Paz Contreras, Equipo de enfermería de la Universidad de Los Lagos Osorno, Catherine Fernández, Camila Rojas, Paulina Lagos, Rocío Mejías, Melissa Canales, Patricio Suazo, Pamela Ángel, Romina Inostroza, Pamela Silva, Felipe Collado, Vanina Cuevas, Rodrigo Oñate, Daniel Salamanca, Javier Briones, Vanessa Villagrán, Diana Bocaz, Andrés Umanzor.

This work was supported by the National Agency of Research and Development (ANID) projects COVID0422 and FONDEF ID20I10192. We also thank USS-FIN-23-PDOC-02.

References

- Ahern D. J., Ai Z., Ainsworth M., Allan C., Allcock A., Angus B., Zurke (2022) **A blood atlas of COVID-19 defines hallmarks of disease severity and specificity** *Cell* **185**:916–938 <https://doi.org/10.1016/j.cell.2022.01.012>
- Andrews S. (2010) **Andrews, S. (2010). FastQC: A quality control tool for high throughput sequence data. Retrieved from** <https://www.bioinformatics.babraham.ac.uk/projects/fastqc/>
- Babatunde K. A., Adenuga O. F (2022) **Neutrophils in malaria: A double-edged sword role** *Frontiers in immunology* **13** <https://doi.org/10.3389/fimmu.2022.922377>
- Bastard P., Rosen L. B., Zhang Q., Michailidis E., Hoffmann H.-H., Zhang Y., Casanova (2020) **Autoantibodies against type I IFNs in patients with life-threatening COVID-19** *Science* **370** <https://doi.org/10.1126/science.abd4585>
- Bernal E., Gimeno L., Alcaraz M. J., Quadeer A. A., Moreno M., Martínez-Sánchez M. V, on Behalf of Murcian COVID19 Study Group (2021) **Activating Killer-cell Immunoglobulin-like Receptors are associated with the severity of COVID-19** *The Journal of Infectious Diseases* <https://doi.org/10.1093/infdis/jiab228>
- Bernardes J. P., Mishra N., Tran F., Bahmer T., Best L., Blase J. I., Ziebuhr (2020) **Longitudinal Multi-omics Analyses Identify Responses of Megakaryocytes Erythroid Cells, and Plasmablasts as Hallmarks of Severe COVID-19.** *Immunity* **53** <https://doi.org/10.1016/j.immuni.2020.11.017>
- Björkström N. K., Strunz B., Ljunggren H.-G (2022) **Natural killer cells in antiviral immunity** *Nature Reviews Immunology* **22**:112–123 <https://doi.org/10.1038/s41577-021-00558-3>
- Borrego F., Masilamani M., Kabat J., Sanni T. B., Coligan J. E (2005) **The cell biology of the human natural killer cell CD94/NKG2A inhibitory receptor** *Molecular Immunology* **42**:485–488 <https://doi.org/10.1016/j.molimm.2004.07.031>
- Borrego F., Masilamani M., Marusina A. I., Tang X., Coligan J. E (2006) **The CD94/NKG2 family of receptors** *Immunologic Research* **35**:263–277 <https://doi.org/10.1385/IR:35:3:263>
- Casado J. L., Moraga E., Vizcarra P., Velasco H., Martín-Hondarza A., Haemmerle J., Vallejo (2022) **Expansion of CD56dimCD16neg NK Cell Subset and Increased Inhibitory KIRs in Hospitalized COVID-19 Patients** *Viruses* **14**
- Chen Z., John Wherry E (2020) **T cell responses in patients with COVID-19** *Nature Reviews Immunology* **20**:529–536 <https://doi.org/10.1038/s41577-020-0402-6>
- Daamen A. R., Bachali P., Owen K. A., Kingsmore K. M., Hubbard E. L., Labonte A. C., Lipsky P., E (2021) **Comprehensive transcriptomic analysis of COVID-19 blood, lung, and airway** *Scientific Reports* **11** <https://doi.org/10.1038/s41598-021-86002-x>

Delorey T. M., Ziegler C. G. K., Heimberg G., Normand R., Yang Y., Segerstolpe Å., Regev (2021) **COVID-19 tissue atlases reveal SARS-CoV-2 pathology and cellular targets** *Nature* **595**:107–113 <https://doi.org/10.1038/s41586-021-03570-8>

Fang M., Orr Mark T., Spee P., Egebjerg T., Lanier Lewis L., Sigal Luis J (2011) **CD94 Is Essential for NK Cell-Mediated Resistance to a Lethal Viral Disease** *Immunity* **34**:579–589 <https://doi.org/10.1016/j.immuni.2011.02.015>

Fielding C. A., Sabberwal P., Williamson J. C., Greenwood E. J. D., Crozier T. W. M., Zelek W, Stanton R. J (2022) **SARS-CoV-2 host-shutoff impacts innate NK cell functions, but antibody-dependent NK activity is strongly activated through non-spike antibodies** *eLife* **11** <https://doi.org/10.7554/eLife.74489>

Gardiner C. M (2008) **Killer cell immunoglobulin-like receptors on NK cells: the how, where and why** *International Journal of Immunogenetics* **35**:1–8 <https://doi.org/10.1111/j.1744-313X.2007.00739.x>

Ge H., Wang X., Yuan X., Xiao G., Wang C., Deng T., Xiao (2020) **The epidemiology and clinical information about COVID-19** *European Journal of Clinical Microbiology & Infectious Diseases* **39**:1011–1019 <https://doi.org/10.1007/s10096-020-03874-z>

Haitao T., Vermunt J. V., Abeykoon J., Ghamrawi R., Gunaratne M., Jayachandran M., Garovic V., D (2020) **COVID-19 and Sex Differences: Mechanisms and Biomarkers** *Mayo Clinic Proceedings* **95**:2189–2203 <https://doi.org/10.1016/j.mayocp.2020.07.024>

Janssen N. A. F., Grondman I., de Nooijer A. H., Boahen C. K., Koeken V., Matzaraki V., van de Veerdonk F., L (2021) **Dysregulated Innate and Adaptive Immune Responses Discriminate Disease Severity in COVID-19** *J Infect Dis* **223**:1322–1333 <https://doi.org/10.1093/infdis/jiab065>

Jevtic S. D., Nazy I (2022) **The COVID Complex: A Review of Platelet Activation and Immune Complexes in COVID-19** *Frontiers in immunology* **13** <https://doi.org/10.3389/fimmu.2022.807934>

Kang S., Brown H. M., Hwang S (2018) **Direct Antiviral Mechanisms of Interferon-Gamma** *Immune Netw* **18** <https://doi.org/10.4110/in.2018.18.e33>

Khalil B. A., Elemam N. M., Maghazachi A. A (2021) **Chemokines and chemokine receptors during COVID-19 infection** *Computational and Structural Biotechnology Journal* **19**:976–988 <https://doi.org/10.1016/j.csbj.2021.01.034>

Krueger F **Krueger, F. Trim Galore! Retrieved from** https://www.bioinformatics.babraham.ac.uk/projects/trim_galore/

Ladds E., Rushforth A., Wieringa S., Taylor S., Rayner C., Husain L., Greenhalgh T (2020) **Persistent symptoms after Covid-19: qualitative study of 114 “long Covid” patients and draft quality principles for services** *BMC Health Serv Res* **20** <https://doi.org/10.1186/s12913-020-06001-y>

Langfelder P., Horvath S (2008) **WGCNA: an R package for weighted correlation network analysis** *BMC Bioinformatics* **9** <https://doi.org/10.1186/1471-2105-9-559>

Lee M. J., Leong M. W., Rustagi A., Beck A., Zeng L., Holmes S., Blish C., A (2022) **SARS-CoV-2 escapes direct NK cell killing through Nsp1-mediated downregulation of ligands for NKG2D** *Cell Reports* **41** <https://doi.org/10.1016/j.celrep.2022.111892>

- Lee N., Llano M., Carretero M., Ishitani A., Navarro F., López-Botet M., Geraghty D. E (1998) **HLA-E is a major ligand for the natural killer inhibitory receptor CD94/NKG2A** *Proceedings of the National Academy of Sciences* **95**:5199–5204 <https://doi.org/10.1073/pnas.95.9.5199>
- Lee S. Y., Lee S. H., Seo H. B., Ryu J. G., Jung K., Choi J. W., Cho M., L (2019) **Inhibition of IL-17 ameliorates systemic lupus erythematosus in Roquin(san/san) mice through regulating the balance of TFH cells, GC B cells, Treg and Breg** *Sci Rep* **9** <https://doi.org/10.1038/s41598-019-41534-1>
- Lee Y. K., Mukasa R., Hatton R. D., Weaver C. T (2009) **Developmental plasticity of Th17 and Treg cells** *Current Opinion in Immunology* **21**:274–280 <https://doi.org/10.1016/j.coi.2009.05.021>
- Li X., Zhong X., Wang Y., Zeng X., Luo T., Liu Q (2021) **Clinical determinants of the severity of COVID-19: A systematic review and meta-analysis** *PLOS ONE* **16** <https://doi.org/10.1371/journal.pone.0250602>
- Li Y., Chen S., Li X., Wang X., Li H., Ning S., Chen H (2021) **CD247, a Potential T Cell-Derived Disease Severity and Prognostic Biomarker in Patients With Idiopathic Pulmonary Fibrosis** *Frontiers in immunology* **12** <https://doi.org/10.3389/fimmu.2021.762594>
- Litvinov R. I., Evtugina N. G., Peshkova A. D., Safiullina S. I., Andrianova I. A., Khabirova A. I., Weisel J., W (2021) **Altered platelet and coagulation function in moderate-to-severe COVID-19** *Scientific Reports* **11** <https://doi.org/10.1038/s41598-021-95397-6>
- Mann E. R., Menon M., Knight S. B., Konkell J. E., Jagger C., Shaw T. N., Hussell (2020) **Longitudinal immune profiling reveals key myeloid signatures associated with COVID-19** *Science Immunology* **5** <https://doi.org/10.1126/sciimmunol.abd6197>
- Masood K. I., Yameen M., Ashraf J., Shahid S., Mahmood S. F., Nasir A., Hasan (2021) **Upregulated type I interferon responses in asymptomatic COVID-19 infection are associated with improved clinical outcome** *Scientific Reports* **11** <https://doi.org/10.1038/s41598-021-02489-4>
- Masselli E., Vaccarezza M., Carubbi C., Pozzi G., Presta V., Mirandola P., Vitale M (2020) **NK cells: A double edge sword against SARS-CoV-2** *Advances in Biological Regulation* **77** <https://doi.org/10.1016/j.jbior.2020.100737>
- Maucourant C., Filipovic I., Ponzetta A., Aleman S., Cornillet M., Hertwig L., Karolinska C.-S., G (2020) **Natural killer cell immunotypes related to COVID-19 disease severity** *Sci Immunol* **5** <https://doi.org/10.1126/sciimmunol.abd6832>
- Nalbandian A., Sehgal K., Gupta A., Madhavan M. V., McGroder C., Stevens J. S., Wan E., Y (2021) **Post-acute COVID-19 syndrome** *Nat Med* **27**:601–615 <https://doi.org/10.1038/s41591-021-01283-z>
- Newton A. H., Cardani A., Braciale T. J (2016) **The host immune response in respiratory virus infection: balancing virus clearance and immunopathology** *Seminars in Immunopathology* **38**:471–482 <https://doi.org/10.1007/s00281-016-0558-0>
- Notarbartolo S., Ranzani V., Bandera A., Gruarin P., Bevilacqua V., Putignano A. R., Abrignani (2021) **Integrated longitudinal immunophenotypic, transcriptional and repertoire analyses delineate immune responses in COVID-19 patients** *Science Immunology* **6** <https://doi.org/10.1126/sciimmunol.abg5021>

- Pacha O., Sallman M. A., Evans S. E (2020) **COVID-19: a case for inhibiting IL-17?** *Nature Reviews Immunology* **20**:345–346 <https://doi.org/10.1038/s41577-020-0328-z>
- Paludan S. R., Mogensen T. H (2022) **Innate immunological pathways in COVID-19 pathogenesis** *Science Immunology* **7** <https://doi.org/10.1126/sciimmunol.abm5505>
- Park M. D (2020) **Immune evasion via SARS-CoV-2 ORF8 protein?** *Nature Reviews Immunology* **20**:408–408 <https://doi.org/10.1038/s41577-020-0360-z>
- Passelli K., Billion O., Tacchini-Cottier F (2021) **The Impact of Neutrophil Recruitment to the Skin on the Pathology Induced by Leishmania Infection** *Frontiers in immunology* **12** <https://doi.org/10.3389/fimmu.2021.649348>
- Ponti G., Maccaferri M., Ruini C., Tomasi A., Ozben T (2020) **Biomarkers associated with COVID-19 disease progression** *Critical Reviews in Clinical Laboratory Sciences* **57**:389–399 <https://doi.org/10.1080/10408363.2020.1770685>
- Prager I., Watzl C (2019) **Mechanisms of natural killer cell-mediated cellular cytotoxicity** *Journal of Leukocyte Biology* **105**:1319–1329 <https://doi.org/10.1002/JLB.MR0718-269R>
- Rabaan A. A., Smajlovic S., Tombuloglu H., Cordic S., Hajdarevic A., Kudic N., Al-Suhaimi (2022) **SARS-CoV-2 infection and multi-organ system damage: a review** *Bosn J Basic Med Sci* <https://doi.org/10.17305/bjbms.2022.7762>
- Rangachari M., Mauermann N., Marty R. R., Dirnhofer S., Kurrer M. O., Komnenovic V., Eriksson U. (2006) **T-bet negatively regulates autoimmune myocarditis by suppressing local production of interleukin 17** *Journal of Experimental Medicine* **203**:2009–2019 <https://doi.org/10.1084/jem.20052222>
- Rashid F., Xie Z., Suleman M., Shah A., Khan S., Luo S (2022) **Roles and functions of SARS-CoV-2 proteins in host immune evasion** *Frontiers in immunology* **13** <https://doi.org/10.3389/fimmu.2022.940756>
- Ren X., Wen W., Fan X., Hou W., Su B., Cai P., Zhang (2021) **COVID-19 immune features revealed by a large-scale single-cell transcriptome atlas** *Cell* **184**:1895–1913 <https://doi.org/10.1016/j.cell.2021.01.053>
- Ryan F. J., Hope C. M., Masavuli M. G., Lynn M. A., Mekonnen Z. A., Yeow A. E. L., Lynn D., J (2022) **Long-term perturbation of the peripheral immune system months after SARS-CoV-2 infection** *BMC Medicine* **20** <https://doi.org/10.1186/s12916-021-02228-6>
- M S.B, Chacko J., Selvarajan B., Peter S., Geevar J. V., Dave T., G R., Srivastava (2024) **Biomarkers of coagulation, endothelial, platelet function, and fibrinolysis in patients with COVID-19: a prospective study** *Scientific Reports* **14** <https://doi.org/10.1038/s41598-024-51908-9>
- Sadeghi A., Tahmasebi S., Mahmood A., Kuznetsova M., Valizadeh H., Taghizadieh A., Ahmadi (2021) **Th17 and Treg cells function in SARS-CoV2 patients compared with healthy controls** *Journal of Cellular Physiology* **236**:2829–2839 <https://doi.org/10.1002/jcp.30047>
- Samadizadeh S., Masoudi M., Rastegar M., Salimi V., Shahbaz M. B., Tahamtan A (2021) **COVID-19: Why does disease severity vary among individuals?** *Respir Med* **180** <https://doi.org/10.1016/j.rmed.2021.106356>

- Schultze J. L., Aschenbrenner A. C (2021) **COVID-19 and the human innate immune system** *Cell* **184**:1671–1692 <https://doi.org/10.1016/j.cell.2021.02.029>
- Sette A., Crotty S (2021) **Adaptive immunity to SARS-CoV-2 and COVID-19** *Cell* <https://doi.org/10.1016/j.cell.2021.01.007>
- Shen J., Fan J., Zhao Y., Jiang D., Niu Z., Zhang Z., Cao G (2023) **Innate and adaptive immunity to SARS-CoV-2 and predisposing factors** *Frontiers in immunology* **14** <https://doi.org/10.3389/fimmu.2023.1159326>
- Subramanian A., Nirantharakumar K., Hughes S., Myles P., Williams T., Gokhale K. M., Haroon (2022) **Symptoms and risk factors for long COVID in non-hospitalized adults** *Nat Med* **28**:1706–1714 <https://doi.org/10.1038/s41591-022-01909-w>
- Sun C., Xie C., Bu G.-L., Zhong L.-Y., Zeng M.-S (2022) **Molecular characteristics, immune evasion, and impact of SARS-CoV-2 variants** *Signal Transduction and Targeted Therapy* **7** <https://doi.org/10.1038/s41392-022-01039-2>
- Szklarczyk D., Gable A. L., Nastou K. C., Lyon D., Kirsch R., Pyysalo S., von Mering (2020) **The STRING database in 2021: customizable protein–protein networks, and functional characterization of user-uploaded gene/measurement sets** *Nucleic Acids Research* **49**:D605–D612 <https://doi.org/10.1093/nar/gkaa1074>
- The C.-H. G. I (2020) **The COVID-19 Host Genetics Initiative, a global initiative to elucidate the role of host genetic factors in susceptibility and severity of the SARS-CoV-2 virus pandemic** *European Journal of Human Genetics* **28**:715–718 <https://doi.org/10.1038/s41431-020-0636-6>
- Thorne L. G., Bouhaddou M., Reuschl A.-K., Zuliani-Alvarez L., Polacco B., Pelin A., Krogan N., J (2022) **Evolution of enhanced innate immune evasion by SARS-CoV-2** *Nature* **602**:487–495 <https://doi.org/10.1038/s41586-021-04352-y>
- Uzhachenko R. V., Shanker A (2019) **CD8+ T Lymphocyte and NK Cell Network: Circuitry in the Cytotoxic Domain of Immunity** *Frontiers in immunology* **10** <https://doi.org/10.3389/fimmu.2019.01906>
- Vabret N., Britton G. J., Gruber C., Hegde S., Kim J., Kuksin M., Laserson (2020) **Immunology of COVID-19: Current State of the Science** *Immunity* **52**:910–941 <https://doi.org/10.1016/j.immuni.2020.05.002>
- Varchetta S., Mele D., Oliviero B., Mantovani S., Ludovisi S., Cerino A., Mondelli M., U (2021) **Unique immunological profile in patients with COVID-19** *Cellular & Molecular Immunology* **18**:604–612 <https://doi.org/10.1038/s41423-020-00557-9>
- Vietzen H., Zoufaly A., Traugott M., Aberle J., Aberle S. W., Puchhammer-Stöckl E (2021) **Deletion of the NKG2C receptor encoding KLRC2 gene and HLA-E variants are risk factors for severe COVID-19** *Genetics in Medicine* **23**:963–967 <https://doi.org/10.1038/s41436-020-01077-7>
- Vivier E., Tomasello E., Baratin M., Walzer T., Ugolini S (2008) **Functions of natural killer cells** *Nature Immunology* **9**:503–510 <https://doi.org/10.1038/ni1582>

- Wang C., Huan T., Zhou X., Zhang X., Duan S., Gao S., Li (2022) **Development of Single-Cell Transcriptomics and Its Application in COVID-19** *Viruses* **14** <https://doi.org/10.3390/v14102271>
- Wang J., Li Q., Yin Y., Zhang Y., Cao Y., Lin X., Qiu (2020) **Excessive Neutrophils and Neutrophil Extracellular Traps in COVID-19** *Frontiers in immunology* **11** <https://doi.org/10.3389/fimmu.2020.02063>
- Wauters E., Van Mol P., Garg A. D., Jansen S., Van Herck Y., Vanderbeke L., collaborators (2021) **Discriminating mild from critical COVID-19 by innate and adaptive immune single-cell profiling of bronchoalveolar lavages** *Cell Research* **31**:272–290 <https://doi.org/10.1038/s41422-020-00455-9>
- Wu Z., McGoogan J. M (2020) **Characteristics of and Important Lessons From the Coronavirus Disease 2019 (COVID-19) Outbreak in China: Summary of a Report of 72 314 Cases From the Chinese Center for Disease Control and Prevention** *JAMA* **323**:1239–1242 <https://doi.org/10.1001/jama.2020.2648>
- Xiong Y., Liu Y., Cao L., Wang D., Guo M., Jiang A., Chen (2020) **Transcriptomic characteristics of bronchoalveolar lavage fluid and peripheral blood mononuclear cells in COVID-19 patients** *Emerging microbes & infections* **9**:761–770 <https://doi.org/10.1080/22221751.2020.1747363>
- Yu G., Wang L.-G., Han Y., He Q.-Y (2012) **clusterProfiler: an R Package for Comparing Biological Themes Among Gene Clusters** *OMICS: A Journal of Integrative Biology* **16**:284–287 <https://doi.org/10.1089/omi.2011.0118>
- Zhang B., Zhou X., Zhu C., Song Y., Feng F., Qiu Y., Wang (2020) **Immune Phenotyping Based on the Neutrophil-to-Lymphocyte Ratio and IgG Level Predicts Disease Severity and Outcome for Patients With COVID-19** *Frontiers in Molecular Biosciences* **7** <https://doi.org/10.3389/fmolb.2020.00157>
- Zhang L., Guo H (2020) **Biomarkers of COVID-19 and technologies to combat SARS-CoV-2** *Advances in Biomarker Sciences and Technology* **2**:1–23 <https://doi.org/10.1016/j.abst.2020.08.001>
- Zhang S., Cooper-Knock J., Weimer A. K., Shi M., Kozhaya L., Unutmaz D., Snyder M., P (2022) **Multomic analysis reveals cell-type-specific molecular determinants of COVID-19 severity** *Cell Systems* <https://doi.org/10.1016/j.cels.2022.05.007>
- Zheng H.-Y., Xu M., Yang C.-X., Tian R.-R., Zhang M., Li J.-J., Zheng (2020) **Longitudinal transcriptome analyses show robust T cell immunity during recovery from COVID-19** *Signal Transduction and Targeted Therapy* **5** <https://doi.org/10.1038/s41392-020-00457-4>
- Zingoni A., Molfetta R., Fionda C., Soriani A., Paolini R., Cipitelli M., Santoni A. (2018) **NKG2D and Its Ligands: “One for All, All for One”** *Frontiers in immunology* **9** <https://doi.org/10.3389/fimmu.2018.00476>
- Zuo Y., Yalavarthi S., Shi H., Gockman K., Zuo M., Madison J. A., Knight J., S (2020) **Neutrophil extracellular traps in COVID-19** *JCI Insight* **5** <https://doi.org/10.1172/jci.insight.138999>

Editors

Reviewing Editor

Wayne Yokoyama

Washington University in St. Louis, St Louis, United States of America

Senior Editor

Betty Diamond

The Feinstein Institute for Medical Research, Manhasset, United States of America

Reviewer #1 (Public Review):

Summary:

Medina et al, 2023 investigated the peripheral blood transcriptional responses in patients with diversifying disease outcomes. The authors characterized the blood transcriptome of four non-hospitalized individuals presenting mild disease and four patients hospitalized with severe disease. These individuals were observed longitudinally at three timepoints (0-, 7-, and 28-days post recruitment), and distinct transcriptional responses were observed between severe hospitalized patients and mild non-hospitalized individuals, especially during 0- and 7-day collection timepoints. Particularly, the authors found that increased expression of genes associated with NK cell cytotoxicity is associated with mild outcomes. Additional co-regulated gene network analyses positively correlates T cell activity with mild disease and neutrophil degranulation with severe disease.

Strengths:

The longitudinal measurements in individual participants at consistent collection intervals can offer an added dimension to the dataset that involves temporal trajectories of genes associated with disease outcomes and is a key strength of the study. The use of co-expressed gene networks specific to the cohort to complement enrichment results obtained from pre-determined gene sets can offer valuable insights into new associations/networks associated with disease progression and warrants further analyses on the biological functions enriched within these co-expressed network modules.

Weaknesses:

There is a large difference in the infection timeline (onset of symptom to recruitment) between mild and severe patient cohort. As immune responses during early infection can be highly dynamic, the differences in infection timeline may bias transcriptional signatures observed between the groups. The study is also limited by a small cohort size.

Comments on revised version:

The authors have addressed the specific concerns brought forth by the reviewers.

<https://doi.org/10.7554/eLife.94242.2.sa2>

Reviewer #2 (Public Review):

In their manuscript, Medina and colleagues investigate transcriptional differences between mild and severe SARS-CoV-2 infections. Their analyses are very comprehensive incorporating a multitude of bioinformatics tools ranging from PCA plots, GSEA and DEG analysis, protein-protein interaction network, and weighted correlation network analyses. They conclude that in mild COVID-19 infection NK cell functionality is compromised and this is connected to

cytokine interactions and Th1/Th2 cell differentiation pathways cross-talk, bridging the innate and the adaptive arms of the immune system. The authors successfully recruited participants with both mild and severe COVID-19 between November 2020 to May 2021. The analyzed cohort is gender and acceptably age-matched and the results reported are promising. Signatures associated with NK cell cytotoxicity in mild and neutrophil functions in the severe group during acute infection are the chief findings reported in this manuscript.

Comments on revised version:

The authors responded appropriately to the previous review critiques.

<https://doi.org/10.7554/eLife.94242.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Recommendations For The Authors):

(1) Due to the significant difference between the infection timeline of mild (1 day post symptom onset) and severe (10 days post symptom onset) cohort at enrollment, an informative analysis to consider is to compare timepoint 2 from the mild cohort to timepoint 1 from the severe cohort.

In agreement with what the reviewer noted on his comment, to be more helpful we completed the analysis comparing timepoint 2 from the mild cohort to timepoint 1 from severe cohort, which is now included as Figure 4-figure supplement 5. The new text added is on pages 13-14, lines 346-355 explaining this analysis. We also included a paragraph in the discussion on page 22, lines 595-604. We have resolved to show this comparison to enforce the main observation related to Natural Killer Cytotoxicity pathways enriched in all analyses of this work.

(2) Alternatively, as this information is available, the authors may group the samples based on the individual's infection timeline as opposed to the recruitment timeline.

Patients in both groups were enrolled at the peak of their symptoms. According to this criterion, we grouped the patients to generate more significant results. Since these infections occurred naturally, we have no accurate information regarding the infection timing of patients. However, if the samples were grouped in order of individual infection timeline, the analysis would be statistically weak to make conclusions about the course of COVID-19, as disease progression would not be coordinated. Our grouping approach provided us a good confidence range, despite the tiny population evaluated.

(3) The authors selected three co-regulated network modules based on the size of module membership genes, selecting the three modules containing the largest gene membership. Small co-regulated networks can also offer important biological insights into specific molecular machinery associated with disease outcomes.

Figure 5 was updated including two more networks (besides blue), for brown and turquoise modules (5E and 5F). This new information allowed us to understand deeply the three larger modules with the most significant results, due to the number of genes they included (blue: 704, brown: 508, and turquoise: 712). The new text describing this analysis is included in page 15 lines 388-396. The remaining 7 modules were also analyzed, and the Gene Ontology/Pathways enrichment were included in 2 new supplemental figures (Figure 5 -

figure supplement 1 and 2). The new text describing this analysis is included on page 15, lines 397-401.

(4) An alternative selection criterion that can inform biological associations between module genes and disease severity is the strength of the correlation coefficients. It seems from Figure 5B, that yellow, turquoise, and green modules have a moderate positive correlation with severe patients, while brown, blue, and gray modules show a slight positive correlation with mild outpatients. A recommendation for the authors is to consider revising Figure 5C to include the enrichment of these additional modules and include these modules in the interpretation of the results.

The correlations between cohorts and the modules (blue, brown and turquoise) are clearly identified for severe or mild patients. However, for several smaller modules, correlations are heterogeneous for different patients of the cohorts, making it hard to gain a clear conclusion related to severity groups. In this sense, the 7 modules were analyzed as is indicated in the previous response number #3, and the results offer an idea of the different transcriptional programs present at different patients in different stages of disease. However, the small number of genes in some modules brings weak results of GO and enriched pathways, making it difficult to interpretation. The text describing this figure is included in page 15 lines 397-401. Also, the network analyses for brown and turquoise modules were included in figure 5 as 5E-F and the text detailing these figures was included on page 15 lines 388-396.

(5) In Figures 3E and 3F, the authors present enrichment analyses of differentially expressed genes from day 28. However, earlier in the results (lines 226-228), the authors reported no differentially expressed genes observed between the mild and severe participant cohort at this time point. Can the authors clarify which comparison was performed to obtain the list of differentially expressed genes used in the enrichment analyses in Figures 3E and 3F?

The discrepancy in this case stems from separate criteria employed for comparison in each case. At the pairwise comparison, DEGs list is different from the longitudinal comparison mentioned afterwards, as for this later analysis we selected only the genes with different trajectories throughout the study (Figure 3). To clarify this point, we included a new paragraph on page 11, lines 278-285.

Original:

“We detected 828 genes that exhibited temporal and quantitative expression level differences during the progression of disease. We discovered additional biological processes and KEGG pathways that were differentially enriched during the COVID-19 progression in mild and severe patients (Figure 3) using the Enrichr platform (G. Chen et al., 2020)”

Changed to:

“To do so, we first identified genes that were differentially expressed between severity groups, and second, we chose only those that also showed changes in their trajectories across sampling times. In doing so, we found 828 genes that exhibited temporal differences in expression level during disease progression. Then using the Enrichr platform (G. Chen et al., 2020), we discovered additional biological processes and KEGG pathways that were differentially enriched during the COVID-19 progression in mild and severe patients (Figure 3).”

(6) Additionally, the authors refer to specific enriched genes in Figure 3 (lines 298-302), but Figure 3 only displays the enriched terms. Can the authors include the results from

the enrichment analysis that include gene membership for each enriched term in the supplement?

Certainly, there is no figure or table in the initial version that includes the gene list for this analysis. We have now included a supplement table 1 and 2 that details each pathway, along with its gene list.

(7) In line 104, can the authors clarify the parameters used to define well-matched samples?

Based on the observations made by the reviewers, we decided to change the wording to make it more obvious about the message of this paper. The update was included on page 5, line as follows:

Original:

“Here, we designed a longitudinal investigation using well-matched samples to study how changes in gene expression in distinct immune effector cells changed during the earliest time points after diagnosis and during progression of clinical disease”,

Changed to:

“Here, we designed a longitudinal comparison between mild and severe patients, choosing the appropriate samples according to the clinical progression and the unbiased gene expression profile”

(8) In lines 113-116, can the authors clarify how their approach mitigates noise/potential biases and very briefly, describe what the nature of noise/biases could be?

The main goal of this paragraph is to show that, while there are several pathways with statistical significance in our analyses, the focus was on NK cell cytotoxicity because this molecular pathway showed bridges between other relevant immune responses; thus, the pathways chosen to respond to its intricate transcriptional program instead of a biased interest. The text was edited and included on page 6, line 111-131 as follows:

Original:

“We used a pairwise comparison of gene expression, gene set enrichment, and weight-correlated gene network analyses to detect differential expression of genes involved with the cytotoxic signaling pathway of Natural Killer (NK) cells in mild versus severe progression of disease. We promoted a broad and integrated point of view throughout the transcriptomic analysis of functional pathways to mitigate noise and potential biases (Bastard et al., 2020; Delorey et al., 2021; Schultze & Aschenbrenner, 2021; S. Zhang et al., 2022). We found close connectivity between NK signaling pathway genes and those of cytokine-cytokine receptor signaling pathways, along with Th1/Th2 cell differentiation genes, as part of the transcriptional circuit executed preferentially among mildly ill patients. Our results detected transcriptional circuits engaging multiple regulatory checkpoints. These findings indicated that the innate NK signaling pathway (cell cytotoxic activity) is beneficial, perhaps a critically-necessary activity needed to effectively eradicate coronavirus. We interpreted that an adaptive immune response that included early cell-mediated immunity was important for reducing disease severity in mild patients. This balance between humoral- and cell-mediated immunity appeared to be less robust in patients presenting with severe COVID-19. These results detected components of the immune response that were significantly associated with the differences in symptom severity observed between mild and severely ill COVID-19 patients.”

Changed to:

“Briefly, to gain more insights into our findings and complement their functional context, we used a pairwise comparison of gene expression, gene set enrichment, and weight-correlated gene network analyses. By doing so, we identified pathways of genes involved with the NK cell cytotoxicity enriched in mild patients when compared to severe. Besides focusing on a particular molecular pathway, we investigated the interactions to better comprehend the underlying phenomena of a successful immune response, contributing to an integrated point of view throughout the transcriptomic analyses of functional pathways to mitigate potential biases attributed to focusing the study on a single pathway. In this regard, we revealed that the NK signaling pathway was intricately related to other transcriptional circuits, such as those governing Th1/Th2 cell differentiation and cytokine-cytokine receptor signaling pathways. These interactions highlight the importance of these pathways as bridges between the innate and adaptive immune responses throughout the disease, implying that the innate NK signaling pathway (cell cytotoxic activity) is beneficial, and possibly a critical activity required to effectively eradicate coronavirus. We also concluded that an adaptive immune response including early cell-mediated immunity was significant in lowering disease severity. The link between the primary innate NK cell activity and the transcriptional priming of adaptive Th1 and Th2 cell responses appears to be more robust in mild patients than in severe.”

(9) In line 120, can the authors clarify which regulatory checkpoints were being referred to?

The concept of “checkpoint” was changed to “bridges” (line 124), because offers a clearer idea about the molecular interaction displayed across the different enriched pathways described in our study. In this sense, the bridges show the connection between innate immune response by NK cell and the adaptive immune response by Th1/Th2 cells

(10) In lines 125-126, can the authors refer to specific results to support this observation?

Lines 111 to 129 summarize the results of the analysis that support the aforementioned phrase. However, the original sentence referred was modified for better comprehension on page 6, lines 129-131 as follows:

Original:

“This balance between humoral- and cell-mediated immunity appeared to be less robust in patients presenting with severe COVID-19”

Changed to:

“The link between the primary innate NK cell activity and the transcriptional priming of adaptive Th1 and Th2 cell responses appears to be more robust in mild patients than in severe.”

(11) In lines 184-185, can the authors clarify what the term “mixed” specifically refers to?

The original text was modified for better comprehension on page 8, lines 177-179 as follows:

Original:

“Interestingly, on day-28, when the majority of patients had recovered, samples from severely ill patients were still mixed compared to those with mild symptoms.”

Changed to:

“Interestingly, on day-28, when the majority of patients had recovered, samples from severely ill patients were pooled together with those mild patients who had already recovered”.

(12) In line 286, can the authors clarify how quantitative expression level differences are distinct from temporal expression level differences?

Despite the differences in the enrollment time between mild and severe cohorts, it was made precisely during COVID-19 symptoms peaks, as illustrated in figure 1B. Also supporting this criterion, the longitudinal analysis outlined in figure 3 was performed taking into account the changes in gene expression trajectories along all sampling times. This point has significance because the results obtained from it exposed several transcriptional programs that were dynamically executed along disease progression, even independently of the pairwise comparison approaches carried out previously.

(13) In Figure 1C, there seemed to be two data points associated with "M1 0 days" and "M4 28 days" with distinct PC projections. Could these samples be mislabeled?

The figure was revised and completed. The hexagon symbol for day-28 was changed for a star symbol. The “M1 0 days” and “M4 28 days” samples were labeled correctly. See below figure 1C with changes as follows:

(14) In Figure 1D caption: could authors clarify if the ranking of 100 genes was based on the log2FC or adjusted p-values?

The criteria considered was Fold Change ≥ 2 and the FDR ≤ 0.05 which is included in the methodology on page 23, lines 657-660

(15) In Figure 4D, can the authors include the expression z score for the healthy participants?

We could include this information, but we consider that it would not help for the understanding of this figure because in this way we put the focus on the differential trajectories between mild and severe patients. Also, DEGs from mild and severe cohorts from this analysis or any other in this work were obtained relatively to healthy donors.

(16) Related to this, can the authors clarify if the expression z scores were computed using the mean and standard deviations of all samples within the study or relative to a specific participant cohort?

The z-score was used considering the mild and severe patients to calculate mean and then the standard deviation of each group. A new paragraph was included in material and methods on page 24, lines 662-664.

(17) In Figure 5B, can the authors include column annotations for participants and sampling time points?

The figure 5B was updated and completed with the suggested information.

(18) In Figure 1 - Figure Supplement 2, can the authors include the volcano plot from the pairwise comparison for day 28 showing no differentially expressed genes between mild and severe participants as reported in the results (lines 226-228)?

The third volcano plot for day 28 was included in the updated figure 1 supplement 2.

Reviewer #2 (Recommendations For The Authors):

The manuscript is generally very well-constructed and well-written. However, the following are the major concerns mostly regarding the study design and participant selection.

(1) The authors have used enrolment day as D0 which is not reflective of the immune response timeline. Especially when the designated 'D0' for the severe group is 10.0 ± 1.8 days post symptom (DPS) onset while the 'D0' for the mild group is 1.2 ± 1.3 DPS. In the context of an acute infection discussed herewith, this difference is critical.

As tempting as it is to conduct longitudinal studies on COVID-19, the authors might do better focusing on specific acute time points (within 10 days post-symptom onset) and convalescent time points (beyond 28 days post-symptom). A better comparison would be D0 severe with D7 mild (aligning the DPS to be between 7-10 days in both groups).

Despite the differences in the enrolment time between mild and severe cohorts, it was made precisely during COVID-19 symptoms peaks, as illustrated in figure 1B. Also supporting this criterion, the longitudinal analysis outlined in figure 3 was performed taking into account the changes in gene expression trajectories along all sampling times. This point has significance because the results obtained from it exposed several transcriptional programs that were dynamically executed along disease progression, even independently of the pairwise comparison approaches carried out previously. Likewise, we agree with the observation of the reviewer, because as we mentioned in the article, it is difficult to properly compare disease progression between naturally infected patients. So, to better support our findings, we complemented them throughout a pairwise comparison between day-7 samples from mild and day-0 samples from severely ill individuals, finding GO terms and enriched pathways related to NK cell function across the mild cohort, as seen in Figure 4-figure supplement 5. This result enforced the main findings gained from the different analyses carried out in this work, highlighting the relevance of the innate immune response of Natural Killer cells, which correlated with a mild progression of disease. The new paragraph describing this analysis was included in pages 13-14, lines 346-355. We also included a paragraph in the discussion on page 22, lines 595-604.

(2) Though there are four participants within each group, one of the participants with severe infection (S1) only has the D0 time point which probably undermines the statistical significance of the results.

This is an accurate observation, as the statistical weight will allow the deeper alterations to be evaluated while the more subtle ones will most likely be excluded from this study. In our analyses, we focused on variations with high statistical significance, which led to the discovery of a distinct Natural Killer response between mild and severe cohorts.

(3) The authors should also account for any medications administered to the severe group in the ICU before enrolment in the study-immune-dampening drugs or steroids which may alter neutrophil recruitment or other immune functions.

Only one severe patient received medication both prior to and during the COVID-19 disease. Even though several medications were administered to this patient, their effects have not been found to increase the neutrophil response.

(4) What was the viral load status at the different time points analyzed - how does this relate to the immune and clinical findings?

In this recruitment the viral load status was not measured.

(5) Was any complete blood count or basic immune phenotyping conducted on these samples? Important to know the various cell frequencies in the PBMC mix sent for sequencing to account for contamination of lymphocytes with RBCs/monocytes/neutrophils as well as any lymphopenia.

This measurement was not done for these samples. However, our protocol of PBMC purification has been tested before and showed small quantities of red blood cell contamination in the process. Furthermore, in all analysis of Gene Ontology or Enriched Pathways, there is not any related to red blood cell genes that could generate noise in the interpretation of our results.

(6) The neutrophil/lymphocyte ratio is already skewed during SARS-CoV-2 infection - which could be the reason for higher readings in severe participants? - speculate?

Effectively, the ratio in several cell types is changed during SARS-CoV-2 infection. However, despite this noise in the proportion of immune cells, different functions in our study are more represented in cells with less count as Natural Killer cells. The modules of co-expression analysis support the notion that despite the number of cells being in different proportions, a transcriptional program is being executed differentially in the cohorts.

(7) CD247/ZAP70 also influences the CD16-mediated NK cell ADCC activity which the authors can add to the innate-adaptive bridging section.

NK CD16a is more highly expressed in NK cells. The circuit involving CD247/ZAP70 and CD16 could explain the cytotoxicity of these cells and how they contribute to the establishment of a response to fight the viral infection of SARS-CoV-2. In our study, CD16a (FcγRIIIa) expression was similar in both mild and severe cohorts. Because our methodology only counts transcriptional changes, genes that did not change were excluded from our discussion. However, our group's research focuses on this node or bridge between innate and adaptive immune responses, with a particular emphasis on fc-antibodies functions, being a topic of interest for future research.

(8) Some of the figures lacked clarity making it difficult to review. (Eg. Fig 4 A, Fig 4 - supplement 1 A&B, Fig 5).

Figure 4A was redesigned, Figure 4-figure supplement 1 was presented in a full page for better resolution.

Specific Comments:

(1) Consider changing "covid-19" in the title of the manuscript to "COVID-19"

Probably the journal platform changes the letters. The original title is in capital letters according to the observation. In the clinical table "COVID-19" was changed to capital letters.

(2) Page 2: Line 24 - Consider revising this line. Not sure what the authors mean by 'early compromise'

The paragraph was revised and rewritten.

Original:

“Mild COVID-19 patients presented an early compromise with NK cell function, whereas severe patients do so with neutrophil function”

Changed to:

”Mild COVID-19 patients displayed an early transcriptional commitment with NK cell function, whereas severe patients do so with neutrophil function”

| (3) Page 4: Lines 57 & 58 - Verify the reference. The paper referenced was published in 2016 and is in regard to SARS-CoV, MERS-CoV, and enterovirus D68.

Effectively, this reference was appropriate for drawing parallels with other respiratory viruses. Due to the emphasis on SARS-CoV-2, the paragraph has been strengthened with two additional references: Shen 2023, and Wauters 2022.

| (4) Page 10: Lines 229 - 234 - Consider referring to the appropriate figure (i.e., Figure Supplement 2 A or B). The figure associated with D28 DEGs (Volcano plot) is missing in the supplementary. Erroneously referred here as Figure 1C which is a PCA plot?

The original text was changed because the figure referenced was correct but misunderstood. The final sentence is on page 9, lines 220-223.

| (5) Page 10: Line 224 - Change the sentence to " We found upregulated.." instead of " We found regulated..".

The text was edited in accordance with this recommendation, which is currently found in line 232.

| (6) Page 13: Line 326 - Figure 4A referenced here is not clear - unable to review.

Figure 4A was updated for a better resolution and included in the manuscript.

| (7) Page 15: Line 398 - Consider rewording "after diagnosis" since the days here are "after enrolment".

This recommendation was considered and the text was rewritten on page 15, lines 404-406:

Original:

“We systematically analyzed transcriptomic features of PBMCs from COVID-19 patients with mild and severe symptoms at three sequential time-points (D0, D7, and D28) after diagnosis”

Changed to:

“We systematically analyzed transcriptomic features of PBMCs from COVID-19 patients with mild and severe symptoms at three sequential time-points (D0, D7, and D28) during the peak of the symptoms”

| (8) Page 17: Move text from the next page to eliminate blank space.

Resolved

| (9) Page 32: Figure 1C - Consider changing the symbol for D28 since it looks very similar to the D0 symbol. Use the colors consistently instead of different shades for each group.

The hexagon symbol was changed by a star symbol for D28 in figure 1C. In this figure each color indicates the three different groups, and the transparent color was used to differentiate the symbols when are close together.

| (10) Page 36: Figure 4A - Unable to review.

This figure was resized for better resolution.

| (11) Page 42-49: Consider relabeling and renumbering the Supplementary figures for consistency and reference the modified numbers in the appropriate location in the main text.

The supplementary figures were relabeling for consistency and better understanding.

| (12) Pages 44 & 48: Unable to review the figures.

The figures indicated were resized for better resolution.

| Examples of consistency review:

| (1) Use of D0,D7 / D-0, D-7 throughout the manuscript

The selected format for the final version of the manuscript is D0, D7, and D28.

| (2) Reporting the source of reagents consistently (Name, Place, Country, Catalog number)

The source reagents were reformatted for consistency in lines 626-628-632-642.

<https://doi.org/10.7554/eLife.94242.2.sa0>